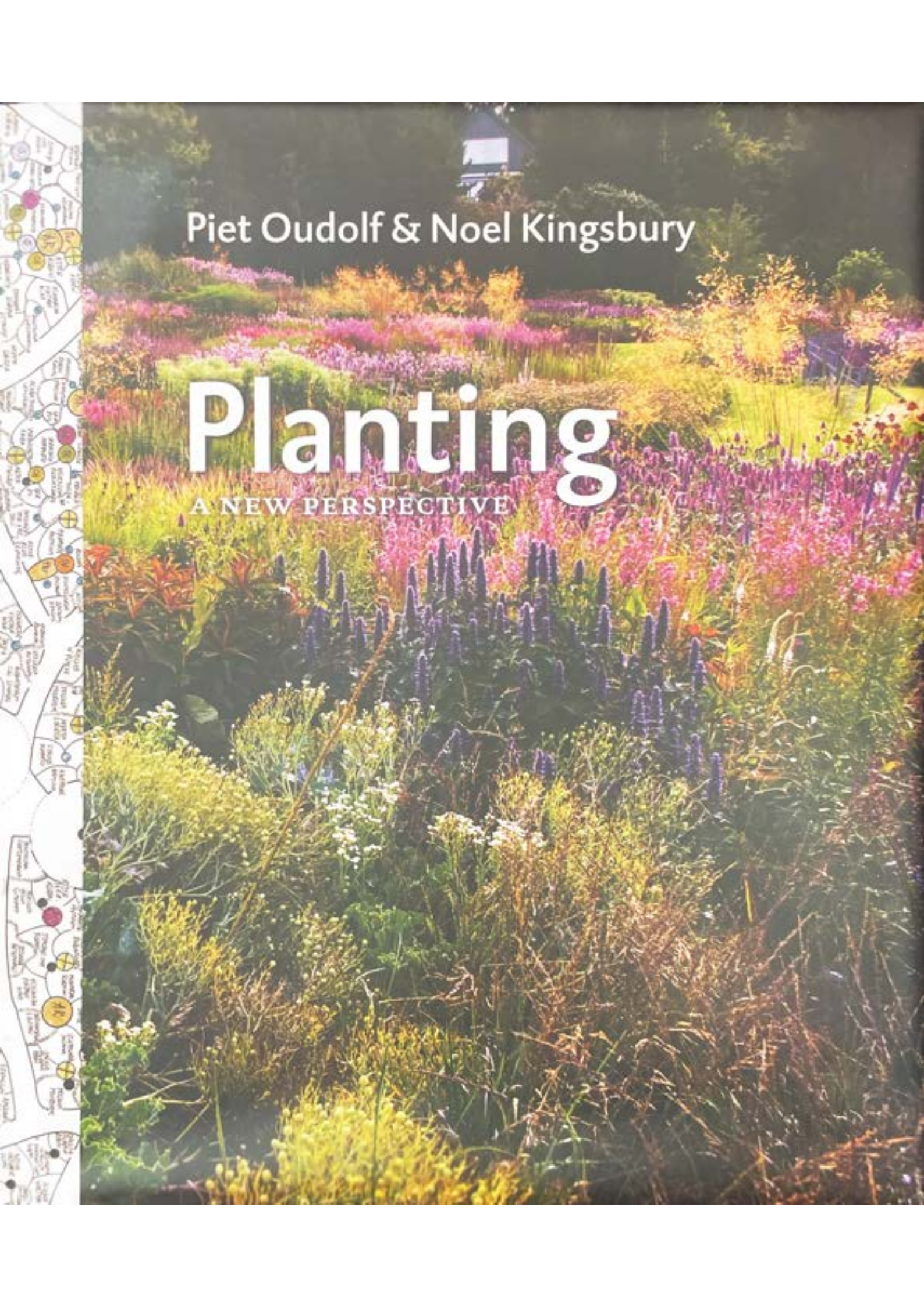




Piet Oudolf & Noel Kingsbury

Planting

A NEW PERSPECTIVE



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►► (Pages 6-7) *Angelica gigas* seedheads stand like spectres amongst a range of late flowering perennials in Piet and Anja Oudolf's garden.

All planting plans discussed and featured in the book can also be studied online. You can find them on www.oudolf.com and www.noelkingsbury.com

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— INTRODUCTION —

PLANTING DESIGN FOR THE TWENTY-FIRST CENTURY

Plants are increasingly being recognized as a vital part of our urban and domestic environments, not just a luxury or an unnecessary – if pleasant – bit of decoration. It has long been established, for example, that the mere view of plants through a window has a beneficial effect on the human psyche, and that plants can play an important role in cleaning and purifying the air of buildings and built-up environments.

Gardening, whether on the most intimate private level or the most extensive and public, involves an appreciation of and involvement with the natural world. For many people, plants may be their only point of contact with nature apart from feeling the effects of the weather. Private gardens offer the opportunity for personal choices to be made about what plants to grow and how to manage them, while designers of civic landscapes have always had the responsibility of serving the wider public interest. There is, however, a new and additional agenda for gardeners, both private and public: sustainability and the support of biodiversity. Sustainability demands that we minimize irreplaceable inputs in gardening and reduce harmful outputs, while the support

of biodiversity brings a demand for wildlife-friendly planting and practices.

The use of long-lived perennials in conjunction with woody plants – the approach Piet Oudolf and I, Noel Kingsbury, have always supported – genuinely offers improved sustainability and support for biodiversity. Reducing the amount of regularly mown lawn and the unnecessary trimming of woody plants for unclear motives is surely a step forward. Creating rich garden habitats offers natural beauty close at hand, provides resources and homes for wildlife, and improves the sustainability of management.

Deciding what plants to use and how to arrange them is covered by the field of planting design, which









◀◀ (page 8)

A green roof around a conference facility: Moorgate Crofts, Rotherham, Yorkshire, designed by Nigel Dunnett (2005). A variable 10–20 centimeter deep substrate supports a wide variety of plant life (around 50 species), selected to maximize the length of the flowering season. This is an example of a semi-intensive green roof, where visual appeal is important as well as functionality.

▲

The former nursery at Piet and Anja Oudolf's at Hummelo in the Netherlands. It is now a boldly experimental area where robust perennials grow amidst a sown mix of wild pasture grasses along with various spontaneously arriving species. Only time will tell how it will work out. The pale mauve-blue is *Aster 'Little Carlow'*, the dark red *Eupatorium maculatum 'Riesenschirm'*.

brings together a combination of technical knowledge and artistic vision. This book looks at some of the recent trends within planting design, and is aimed primarily at home gardeners, garden design and maintenance professionals, and landscape architects. It has important lessons for others, such as architects, who do not use plants directly but often have to situate their work in close proximity to them, or ecologists, whose profession does not involve much design but who increasingly have a role to play in the creation and management of designed plantings.

While the role of plants – and therefore planting design – is well established in the domestic garden, and is indeed absolutely crucial to its aesthetic and functional success, it has not been so well established in landscape design. Or perhaps more accurately, plants have often played a minor role in urban landscape design. Histori-

cally, for centuries the only plants used in cities were avenue trees; the nineteenth century saw the growth of urban parks, the late twentieth a much wider use of plants in urban areas – a practice to a large extent pioneered in the Netherlands. Now, however, the use of plants is increasing, particularly that of perennials and ornamental grasses, requiring greater access to technical information about plant establishment and management, and to ideas about the visual aspects of their use. It is perhaps worth looking at these new trends.

Green roofs

While green roofs have captured the public imagination, the technology involved in their construction is also being used in many other situations, often not immediately recognisable as roofs. Increasingly, high-density urban environments are creating artificial grow-



▲ A randomized perennial mix designed by Nigel Dunnett at Sandvik Tools, Warwickshire (2007), designed to capture and hold water during periods of high rainfall and so prevent flooding. The dark purple is *Aster novi-belgii* 'Purple Dome'.



▲ This swale, designed by Nigel Dunnett for London's Olympic Park, illustrates well a basic concept of sustainable drainage schemes. Drainage water is collected in the lowest part of the site and allowed to drain slowly into the water table, with only the excess being taken off site. A diverse community of British native plant species are used; clearly visible are white *Leucanthemum vulgare* and pink *Lythrum salicaria*.

ing situations which require soil or artificial soil-like substrates above impermeable surfaces. An example of such a green roof which is not a green roof might be Chicago's Lurie Garden – built over a parking garage.

Green roofs are often categorized by the use to which they are put: extensive roofs are functional and usually have a shallow substrate; intensive are more likely to be roof gardens with conventional planting; semi-intensive ones are somewhere between and are developed for their visual appeal as well as functionality. Extensive and semi-intensive roofs tend to be planted up with plant 'communities' – combinations of plants selected for the conditions which coexist well and survive with minimal maintenance, often species from dry meadow habitats. In this sense there is a real distinction between these and conventional plantings, where plants are placed individually.

Water management

A major environmental problem in cities concerns the management of water, especially the control of excessive run-off during storms, which can cause flooding and associated pollution. Sustainable urban drainage schemes (SUDS) are designed to capture water, and either store it or release it slowly into the water table or streams or allow it to be evaporated back into the atmosphere. Domestic gardeners in some cities are encouraged to develop rain gardens where the same objectives are reached, with no drainage water leaving the property and the minimum amount of irrigation water brought in. Green roofs, with their capacity to store rain as it lands, often play a role in integrated SUDS. SUDS often involve swales – detention basins where water can be held temporarily before slow release into natural drainage systems. Often these are planted up with a carefully selected vegetation,

usually locally native species which can survive periodic inundation but dry conditions too. More ornamental planting combinations are, however, also possible. Again, the emphasis is on mixtures of species.

Biofiltration

Biofiltration is the reduction of polluting chemicals in the environment through the use of plants. Street trees, green roofs and living walls are all examples of plant life which can be effective at trapping dust, and breaking down complex hydrocarbons (often known as Volatile Organic Compounds – VOCs) into harmless CO₂ and water. Plants differ in their ability to break down specific VOCs, so combinations are often more effective than just one species.

A more targeted biofiltration technique is in the idea of the natural swimming pool, developed in Germany, where plants act in conjunction with beneficial bacteria to break down or trap nitrogen and phosphorus compounds, so denying nutrients to disease-causing bacteria. The plants used are often grown as a mixed community, as different species have different roles to play in a complex set of biochemical reactions.

Spontaneous vegetation

Post-industrial societies often leave behind large areas of wasteland: abandoned industrial facilities, railways, mines, military training grounds and the like. These are often overrun with plants with remarkable speed, bringing home to us very forcefully how rapidly and thoroughly nature can recover lost ground and heal polluted and damaged environments. Such places often develop a fascinating and unique flora, a combination of locally native species, weeds and escaped garden plants. So often 'redevelopment' means the destruction of these unique plant communities. Recent years have seen a more positive interpretation of these places; led by Germany, areas of post-industrial wasteland are being managed more respectfully and creatively to preserve something of their special character.

One of the most famous examples of such plant conquest of abandoned land is the New York City High Line, an elevated freight railway line last used in the 1960s. The last part of the line was closed in 1980. Its redevelopment as a park had to involve complete rebuilding so the flora was lost, but Piet Oudolf's collaboration

with landscape architects James Corner Field Operations is intended to evoke the wild feeling of the old High Line. The remarkable success of the project is now inspiring other projects in the USA, and highlighting the importance of post-industrial landscapes.

What is striking about many of these new and distinctly technological and engineering approaches to planting is that communities of multiple species are being used – relatively stable groups of plants which can be managed altogether as a unit. This is part of the zeitgeist of contemporary planting design – a slow move away from precise individual plant placement to combinations of species, designing and planting something which is greater than the sum of the parts, developing a vegetation rather than planting a mass of individuals.

Key to the idea of combinations of plants is the intermingling, mixing or blending of species, as opposed to the use of blocks or groups. This creates an effect which is visually more complex and naturalistic. This also means that there is more interaction between plants, and so more competition – as a consequence there is a need for a greater knowledge of ecological issues or at least a greater awareness of long-term performance.

This book aims to explore the new emerging planting design based on intermingling combinations. It is centered around the work of Piet Oudolf, a Dutch landscape designer who works on private and public commissions in northern Europe and North America. It also includes some material about other practitioners working with plant combinations in several different countries: landscape architects, academic researchers and public space managers. I – Noel Kingsbury – have written the text, but this has been very much a joint project between Piet and myself, so for most of this book I am writing on behalf of both of us. Chapter four, 'Long-term Plant Performance', however, has been the result of research as part of a doctoral thesis with the Department of Landscape at Sheffield University followed by some further formal research work; consequently, I have written this with my voice.

What unites the two of us is our passion for and close involvement with plants. Just as the best potters have an intuitive understanding of clays and glazes and master cabinet makers a deep feeling for every type of wood they use, skilled planting designers have a knowl-

edge of the plants that make up their repertoire. Piet's work as a designer is based on 35 years of not just using plants in designs but growing them himself and, from 1982 to 2010, running a nursery with his wife, Anja. I have a background in nursery work myself (albeit for a very much shorter period) – there is nothing like it for gaining an intimate knowledge of how plants behave, what they do and when they do it, and – crucially – what goes on underground: how roots behave. The knowledge needed to propagate plants and then grow them on is that of plant performance – such experience is then invaluable for understanding how they will perform over many years. This surely is a major part of the Oudolf success story.

It is helpful to think about the material in this book as covering two gradients: from macro to micro and from order to spontaneity. The first chapter, 'Planting – The Big Picture', looks at the context of plantings and the gradient from order to spontaneity. The second chapter, 'Grouping Plants', considers the middle level: how plants can be put together in different ways. This is very much about Piet's work, for he is a true designer, in contrast to many professionals whose work is also featured in the book, who could perhaps be better described as ecological engineers. Piet's plants are precisely placed, and indeed much of this chapter is given over to a study of his planting plans. Over time, his plants may move about – which is fine – but the intention is that such changes will only very slowly degrade the original design concept.

Chapter three, 'Combining Plants', comes down to a lower level, looking at combination and juxtaposition: why one plant looks good next to another, how combinations can be made to look good at particular times of year, why one grouping may change with time faster

than another. I pay particular attention to plant 'architecture' – their shape and structure. This chapter will be of most interest and use to the new gardener or designer and to those with only small spaces. Chapter four on performance looks at issues crucial to the long-term survival and spread of perennial plants, but also at death and disappearance. Such issues are vital to understand, not just for the ongoing management of plantings but also at the planning stage.

A fifth chapter covers the work of others who are at the naturalistic cutting-edge in planting design. Whereas Piet's work can be seen as an artistically stylised version of natural habitats, most other practitioners in the field are more concerned with a randomizing of individual plants from a carefully researched selection. The term 'ecological engineers' describes their approach, which combines a technical knowledge of plant performance with visual appeal to create relatively stable and highly decorative plant communities.

Garden and landscape making is increasingly becoming a globalized business. There was a time when the vast majority of ornamental or amenity plant growing happened in the cool temperate climates of north-west Europe, North America, their southerly Mediterranean climate fringes and Japan. Anything done beyond these areas tended to be quite 'colonial' and derivative in character. This is now changing. As the so-called emerging markets generate more resources for public and private garden making, immense new possibilities are opening up. One is for the ongoing development of garden traditions which became stalled – either through cultural decline or the impact of imperialism – notably Islamic, Chinese and Thai. Another is working with plant palettes which have never before been utilized in design.

►► (pages 14–15)

A spontaneous meadow garden at Carrie Furnace, Pittsburgh, Pennsylvania, with spikes of *Echium vulgare* particularly prominent. Post-industrial landscapes can be remarkably biodiverse and even beautiful. Landscape consultant Rick Darke is involved with managing this abandoned industrial

area, describing his role as 'to identify the useful components of the existing spontaneous vegetation on the site and ways of integrating this in the eventual designs... which will have the historic ironworks at the core of mixed-used residential, professional, retail and trailways development.'









This latter approach, the use of novel plants, is tremendously exciting. Much garden design in non-industrial or emerging economies in the past has used a limited flora, repeated all over the world across appropriate climate zones; the monotony of endlessly meeting bougainvilleas, yuccas and *Ficus benjamina* wherever you travel in the tropics is deeply disappointing. The garden, landscape and nursery professions are now increasingly turning to their own floras rather than this global lowest common denominator mix. The reasons are many: a desire to celebrate regional diversity, patriotism, a wish to support biodiversity, and a need for species which are guaranteed to survive the onslaught of difficult local climates.

One example illustrates how this process can work. Amalia Robredo is a garden designer who practices on

the coast of Uruguay, an area with a number of interesting plant communities, now seriously endangered by development. Over the last few years she has systematically made herbarium specimens, got them identified by botanists at the University of Montevideo, collected seed, grown and trialled plants in her garden, persuaded local nurseries to start growing them, and then used them in projects. For a coastal climate, with frequent high-velocity and desiccating winds and sandy soils, the use of such plants makes sense, as well as replacing lost habitat and encouraging people locally to value their environment. For us there will be further advantages, as new species adaptable to other climates make their way into global commerce – not as in the past as species introduced directly from the wild, but already tried and tested by local nurseries and gardeners. Piet's work in



◀◀ (pages 16–17)

The *Salvia* and *Achillea* planting at Hermannshof, Weinheim, in Germany's Rhine Valley, is an example of a naturalistic planting designed for color and structure and to thrive on dry alkaline soils. The blue-purple are seedlings of *Salvia nemorosa* and *S. xsuperba* cultivars, the blue a hybrid of *Veronica longifolia*. The tall yellow is *Verbascum speciosum*, a self-seeding biennial. In the foreground is grass *Festuca mairei* from the Atlas Mountains in north-west Africa, where the climate is similar to regions with steppe or short-grass prairie habitats.



▲

Amalia Robredo in Uruguay is one of the pioneering plant-oriented designers using locally native species for the first time in their regions. The wild flora of the Uruguayan coast has many attractive plant species. Here the grass *Andropogon lindmanii* grows alongside silver *Achyrocline satureioides*.

▲

A green roof on the Uruguayan coast designed by Karina Hogg, with planting by Amalia Robredo uses a mix of native (recently introduced into cultivation) and non-native species. Conditions are extreme with salt spray and exposure; the substrate is 5–7 centimeters. The silver is a native sand dune species, *Senecio crassiflorus*; the yellow daisy is a rare local endemic, *Grindelia orientalis*; the blue is *Plectranthus neochilus*. The grass in the foreground is locally native *Stipa filifolia*.



With the High Line now so popular, many other communities are looking at abandoned rail lines as potential public spaces. Like many former industrial sites, their poor and sometimes polluted substrate, which hardly earns the name of 'soil', is actually very good for encouraging a wide range of plants; it is a counter-intuitive fact of ecology that stressful environments often support more species diversity than fertile ones. Here, in Reading, Pennsylvania, garden escapes *Macleaya cordata* and a species of *Verbascum* are prominent. In other locations rare natives may find a niche.

►► (pages 20–21)

The design for New York City's High Line was intended to evoke the spontaneous vegetation of the old abandoned railway line. Here, in October, are seedheads of the grass *Sporobolus heterolepis* and *Eupatorium maculatum* 'Gateway' alongside the yellow flowers of *Rudbeckia subtomentosa*.

the United States and his collaboration with native plant specialists such as Roy Diblik of Northwind Perennials in Wisconsin have already contributed to a growing interest in North American perennials among garden and landscape professionals in central and northern Europe.

Practitioners in this field tend to be very collegial. We swap ideas, images and plants. We may produce very different results but share many of the fundamental beliefs, which are also being increasingly shared by others beyond our field, both amateur and professional. We are all passionate about wild plants and wild plant communities, and about the need for gardeners and landscape professionals to create environments which support biodiversity as well as nourishing the human

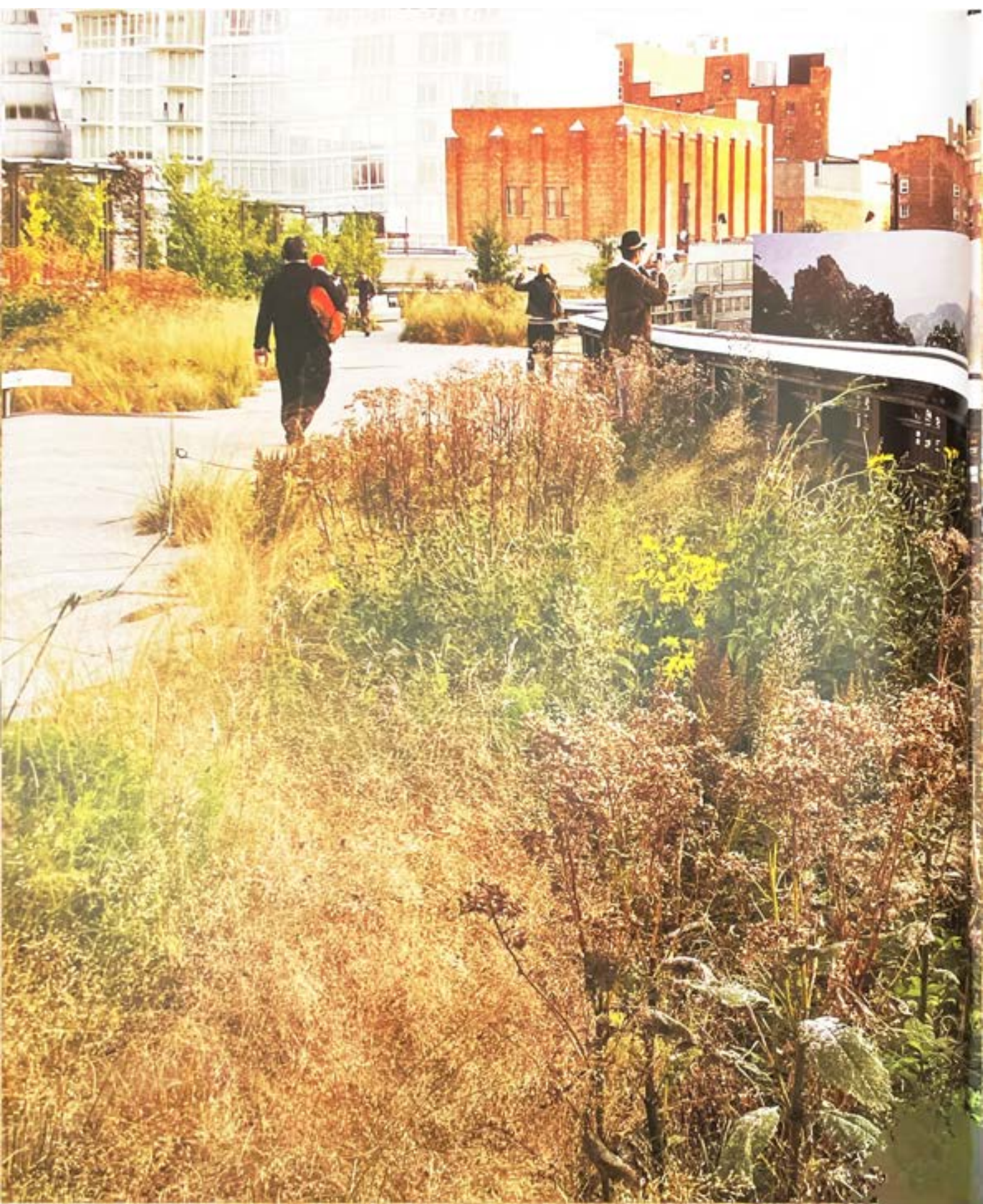
spirit. This book will inspire, encourage and enable others to join us.

We live in interesting times for planting design. There are a great many areas where the integration of plants with architecture and the wider urban fabric is taking plants beyond their merely ornamental or amenity use. Planting design has long been something of a 'Cinderella' within landscape design, and even in the booming profession of garden design it enjoys surprisingly little discussion. Now landscape architects and designers are paying it more attention, and with a growing acceptance of the need for plants to be woven more deeply into the fabric of our lives, I hope this book will make its contribution.

►►► (pages 22–23)

Chicago's Lurie Garden, part of the Millennium Park project, opened in 2004, is an example of a new kind of urban public garden. It is in fact a giant green roof, as it is built over a parking garage, yet reads as part of a ground-level landscape. Offering a stylized version of the Midwest prairie,

its combination of native and introduced perennials and grasses is symbolic of a blend of nature and culture. Such a blend is easily understood by city dwellers, some of whom will be tempted to find out more about wildflowers and so be drawn into wider issues of conservation and biodiversity.











— ONE —

PLANTING – THE BIG PICTURE

Traditionally, the way plants were organized in parks and gardens reflected a culture that liked to order and discipline nature. Contemporary planting design is not only freer, but also seeks to reflect nature. It also addresses our concerns about how we garden sustainably and in partnership with nature.

Even someone who takes the quickest glance at a wild or semi-natural habitat will see that it is a blend of different plant species. Someone who has no interest in wild plants and lives most of their life in a city standing in front of an average field in spring or early summer – by which we mean a traditional low-intensity pasture or hay meadow – will rapidly appreciate that several different kinds of wildflower are present, and that they are all mixed up. Ask them to pay a bit more attention to the grass, and they will soon appreciate that more than one kind can be seen; some in the field form obvious tussocks, others which appear to be spreading on to some bare ground are particularly fine, whereas still others

just seem to form an even-textured mass. The flowering plants present (in Europe, most likely to be meadow buttercup, *Ranunculus acris*, and red clover, *Trifolium pratense*) are scattered all over the field but not evenly – clearly in places the concentration of flowers is greater than in others. Areas of abandoned agricultural or other land can be even better sites to appreciate this complexity of plant life. In much of North America the diversity can often be visually very impressive, particularly later in the growing season, with a range of grasses, wildflowers and wildflower seedheads scattered across a wide area.

◀ Perennials self-seeding at Pensthorpe.

Early spring is a very good time to appreciate the layout of perennial combinations. In the planting below (left) species are in blocks – although most designers would not plant out in straight lines like this (carried out here by the New York City parks authority). The planting below (right) shows a more naturalistic intermingled style (on the High Line) – blocks may still be used, but they are more likely to bleed into each other and some plants are more scattered.



BLOCK OR BLEND?

Looking at a semi-natural grassland is a good way to start thinking about the visual qualities of natural plant communities. Similar patterns occur in woodland, but we famously cannot see the wood for the trees and a good vantage point can be hard to find. Thinking about what we are looking at, it is helpful to focus on the following qualities:

- Intermingling – a wildflower meadow where all the buttercups are neatly organized into one patch, the clover in another and the thistles in another is plainly absurd. Plant communities are very obviously made up of densely intermingled species.
- Diversity – the range of species varies greatly from one sort of community to another, but in most cases there is more diversity than meets the eye. The more

you look, the more you see, particularly in grassland habitats.

- Complexity – imagine trying to count the number of species, let alone the number of individual plants, in a square meter of a field. Part of the problem is the sheer intensity of the intermingled relationships of the plants present.
- Change – complexity is not evenly distributed. Walking through a grassland (North American prairie is the best for this) reveals a constant shift in the distribution of species, or even their presence or absence.
- Coherence – despite the mind-boggling complexity of the plant distribution, there is an overall sense of unity. Standing back so that the whole can be appreciated, the presence of many thousands of individuals is re-

A prairie in flower is a good place to appreciate the patterns of plant distribution. Here several grasses stand out through their form, pale purple *Physostegia virginiana* and cream *Parthenium integrifolium* through their color, and gray *Amarpha canescens* through both color and form. The scattered and intermingled mixing of the species here is very typical of wild plant communities. This is Shoe Factory Road Prairie, Illinois, very near Chicago's O'Hare Airport, in August.

Seedheads of *Echinacea pallida* indicate a loose group of the plant in prairie on a sand and gravel moraine at Glacial Park, Illinois, in September. Such occasional scattering can be dramatic and is easily replicated in cultivation. In the background a colony of *Rhus typhina* can be seen slowly colonizing the grassland – this small tree is used on New York City's High Line, where annual pruning prevents it spreading too much.



solved into something much clearer, the awareness of a ground-mass of greens (or perhaps browns at the end of the year) and a scattering of color (or again at the end of the year, seedhead forms). This simplicity and clarity among complexity gives a sense of coherence. It is perhaps this, more than anything else, which enables us to see real beauty in wild plant communities.

- Distinction – particularly effective at creating a sense of coherence are those species which rise above the mass: not necessarily literally, but in standing out, either through their color, form, height or mass. The European meadow buttercups do this fairly quietly (they achieve their impact through sheer quantity), but other species in different environments may do so more dramatically. The most effective are those which combine color and form/height, such as the magnificent sight of hundreds of spikes of flowers of *Baptisia alba* (white wild indigo) rising above the grasses in an early summer meadow prairie.

MODERNIST MONOCULTURES – PLANTING IN THE TWENTIETH CENTURY

We are all familiar with most cultivated plants being grown in blocks. Arable farming does it on a huge scale, and the sight of a Brueghel painting of medieval peasants harvesting wheat reveals that it has always been so, only the size of the field varying with time and place. A great deal of the landscape planting with which we are familiar is composed of blocks, usually of shrubs, while much of the conventional advice about gardens of the average suburban size was to plant in groups of odd numbers. This convention of monocultural block planting is very much a creation of the twentieth century.

The dominance of the monocultural block as the basic unit of planting in public landscapes and to a lesser (and, needless to say, smaller) extent in private gardens can possibly be seen as a way of simplifying the cost of labor. Much nineteenth-century planting was very

This garden in western Ireland (2006) uses block planting. However, plant groups are repeated so there is a strong sense of rhythm and unity; here, in July, these include *Veronicastrum virginicum* (center), *Sedum* 'Red Cauli' (right foreground), *Molinia* 'Moorhexe' (center foreground) and *Helenium* 'Moerheim Beauty' (right). In addition, clumps of the very tall grass *Stipa gigantea* stand above the rest of the planting and help tie it together.



Repeated single-species blocks of grass *Deschampsia cespitosa* 'Goldtau' create great impact on the large scale at Berne Park, Bottrop, Germany (2010). In the foreground, mixes give more interest; here are grass *Sesleria autumnalis* and *Sedum* 'Matrona'. This is August.



complex, requiring considerable time and skill to design and maintain. More informal planting styles which were developed by designers such as the English Gertrude Jekyll (1843–1932) used drifts – elongated blocks which were soft at the edges and allowed plant partnerships to be seen and appreciated from a variety of angles. Informality and simplification can be seen as the two dominant but linked trends throughout the twentieth century. Organically shaped blocks of perennials in post-war German public planting and much larger blocks with a clear link to mid-century abstract painting in the work of the Brazilian Roberto Burle Marx (1909–1994) illustrate the influence of modernism. Landscape shrubs in blocks in public places were widely used in the Netherlands in the 1960s, a trend in the landscape profession which proved very influential.

Modernism offered a rationale for the simplification and informality of much twentieth-century planting design. It failed to convince everybody, however. From the 1970s onwards, more ecologically focussed gardeners began to promote the use of native plant species and the idea of the garden as a biodiversity preserve – a Dutch group dedicated to these aims took the name *Oase* (oasis). This same period saw the maturing of a movement in Germany dedicated to creating plantings in public spaces through the development of a naturalistic and wildlife-friendly style. This German style, which has come to be known as *Lebensbereich* (habitat), is most chiefly associated with Professor Richard Hansen of Weißenstephan, a complex of trial gardens, research institutes and a university north of Munich; its roots lie in a plant ecology science strongly focussed on the study of plant communities. The most recent developments in this perennial-based planting style are on the creation of randomized planting mixes, which are discussed later in this book.

One plant, one place – advantages and disadvantages of the monocultural block

The single-species block may be dated, but it should not be written off. It has an undeniably simple graphic quality which may often be regarded as valuable, particularly by professionals working in public environments. In private gardens and other intimate settings, its simplicity may be an advantage as a counterpoint to complex perennial-based planting schemes. In fact, its real value

may lie as a counterpoint to the very focus of this book: intermingled plantings may not always be strong on structure, or they may lack structure at certain times of year; the use of firm monocultural blocks may create an exciting creative tension.

In situations where maintenance is limited by lack of time or skill, the advantage of monocultural blocks is very clear, as anything which looks different can be weeded out. This is particularly useful in the case of grasses. Many of us who have been busily promoting naturalistic planting and grasses over the years have been brought up short by situations where carefully chosen specimens are dug out or sprayed with herbicide because inexperienced or badly briefed staff have mistaken them for weed grasses. The unfortunate fact is that, early in the year, even many gardeners who are familiar with grasses can sometimes find it difficult to know which is a desired and which a weed grass. However, by putting grasses into monocultural blocks, it is much easier to avoid such tragedies.

For the most part, however, unless there are clear design, practical or functional reasons for block planting, we would strongly urge gardeners and designers to do their best to move away from this outdated and unimaginative style. In private gardens it simply acts as a deterrent to developing more creative planting schemes, while in the public realm it is little more than a crutch for those in the landscape profession who have for far too long treated plants as green cement. The exception might be the dramatically big block planting, which can have real design impact, but that is another story.

There are more important and more objective arguments against block planting than that it is boring and out of date, however. One is that, like a great actor drunk in the bar after a performance, a whole mass of one plant can look very dreary when it has finished performing. The fact that only a relatively limited number of perennials in the short to medium height range look at all respectable after flowering has meant that a few species tend to be seen again and again. One of the great advantages of Gertrude Jekyll's drifts was that it made it much easier to hold on to a form of block, but by stretching them out she made it easier to hide plants with a 'past the sell-by date' look.





◀◀ (pages 30–31)

The advantage of intermingling plants over block planting is that random close juxtapositions of species are created. As autumn blurs into winter, these will revolve around the silhouettes, textures and forms of plants. Here, in a dry prairie planting at Hermannshof in Germany, grasses (with *Nassella tenuissima* prominent) combine with perennial seedheads (the dark heads are *Echinacea pallida* and *E. paradoxa*). The grayish seedheads to the right are subshrub *Amorpha canescens*.

▶
Chance, or serendipity, should always be allowed to play a part in garden design, and even perhaps in the public landscape too. Here, an unexpected 'meadow' has sprouted over the years in between the paving blocks of a terrace on the south side of the Oudolf family house. Grass *Nassella tenuissima*, a short-lived but strongly seeding species, has germinated alongside purple-blue and pink seedling forms of *Salvia pratensis*, while the questing roots of *Euphorbia cyparissias* have sent up shoots between the pavers.



GETTING THE BLEND RIGHT – THE ADVANTAGES AND DISADVANTAGES OF INTERMINGLING

Moving from planting in blocks to blending, intermingling or mixing – all three words can be used interchangeably – is a major shift in planting design. A working definition would be to say that a 'pure' blend is a complete mixture of individuals of all the species used in a given area. This should not be too much of a challenge for gardeners and designers with good plant knowledge. For some time now, the idea of 'plant combinations' has been an important driver for creativity among both designers and amateur gardeners. A really good combination should have an electric quality about it – two or more plants where a particular complementary relationship of color or form sends the sparks flying. The reality is that these combinations are relatively few. Instead, a great many others are good but not outstanding – particularly useful are those more low-key combinations which look good over a long period. Simply taking a good combination and then repeating it is the

first step in developing an effective intermingled or mixed planting. While two plants together can be dramatic, as a design concept it lacks depth and rarely looks good all year round. Combining four or five plants can create a simple mix, which if chosen well can provide interest for much of the year.

Here, where we discuss intermingled plantings, it is important to emphasize that there are two very different approaches. One is to carefully research a blend and then use it over large areas, either mixing the component plants at random or repeating a designed block, in the way that a mosaicist or tiler laying out a series of small modular units repeats it again and again to create a pattern. There is an analogy with a tea or whisky blender who works intensively on a small quantity of a blend, with an eye to mass production of their eventual choice. Such a planting blend can then be reproduced in two ways, one as literally mass production, with the mix being used anywhere conditions are suitable. Alternatively, it might only be used for one project: a site-specific mix. The other approach is not to randomize or repeat modules but to design in the location of every plant, to

intermingle different combinations in different places, so allowing there to be small groups, repetitions at various frequencies and subtle transition effects.

To summarize, there are three approaches to blended planting here:

- randomized mix, planted or sown
- modular repetition
- designed intermingling.

Of course, there are also 'halfway houses' such as blurring the edges of block planting or randomly scattering different species through a block planting.

Any gardener will quickly be able to suggest that sowing a seed mix is one way to achieve an appearance of randomness. Indeed, this is the approach which James Hitchmough of the University of Sheffield uses. The 'Mixed Planting' and 'Integrated Planting' systems developed in Germany and Switzerland aim to achieve a similar effect by using young plants. Modular repetition is technically difficult to achieve but is possible, and could in theory be used to create some highly effective plant juxtapositions.

Blended plantings are undoubtedly very beautiful and highly effective visually. This alone is an important reason for using them. They also have the advantage that it is possible to easily mix together species which flower or perform at different seasons. They are also forgiving of weed infiltration. As a general rule, the more ordered a planting is, the more problematic weeds are – one weed in a highly geometric bed of summer annuals stands out like a beacon. In wilder, more naturalistic planting styles, occasional weeds stand out less, while intermingled styles have the greatest capacity to absorb intruders. In fact, it is possible to take a pragmatic view and allow some spontaneously occurring species to survive and spread within the planting.

Finally, the advantage of the cost of mixed planting styles, or to be more precise randomized or sown plantings, needs to be considered. One of the reasons for the development of the German and Swiss planting mixes was to reduce the costs of large areas of planting for local government, garden shows and other public sites. The idea is that a mix is developed using careful plant selection, with both practical and aesthetic criteria, and so the design cost is all absorbed at the beginning as a one-off. Anyone can then buy the mix for as many hundred square meters as necessary, with seed or plant costs po-

tentially carrying a percentage royalty for the original design team. The decision of the garden or landscape designer is limited to specifying which mix to use. Design costs are therefore minimized.

It is now time to consider the disadvantages of mixed planting styles. The widespread use of a limited number of planting mixes could rapidly turn them into clichés, much as sedum roofs have become. The fact that commercial nurseries are now marketing planting mixes could well result in a situation where a bold innovation, followed by a mass usage, will then become a boring lowest common denominator. The biodiversity value of such plantings could be good, but to earn popularity with the human element of the landscape – who, after all, select and pay for them – they must also offer visual quality. It is to be hoped that randomized plantings will be an area of continuous research and development, so that new combinations will constantly offer change and novelty. Commercially developed randomized planting mixes may be tailored to particular habitats, but these can only ever be crude approximations to the existing conditions where they are used. They are in no sense 'site-specific'.

There is one other possible problem with randomized mixes, particularly those that are planted rather than sown. This is to do with long-term development and the distribution of species. In nature, plant communities are never random. Certain plant species will colonize an area first, followed by others, so the process of community development is cumulative over time rather than the entire community starting off at once – which is what happens if one is planted. Over time, slight differences in soil and microclimate conditions will cause species to select out, so that there are greater and lesser numbers of different species in different areas. Think about how wildflowers are scattered in a meadow or prairie; part of the visual pleasure is not just the scattering and mingling over a wide area, but the subtle ebb and flow of particular species over space in ever-changing combinations. If species are introduced by sowing, the slight differences in microhabitat will be registered very soon by the developing seedlings, so the mix will grow to fit its landscape. The result will be different trajectories of development in different places, leading to a good ecological fit to the conditions and visual diversity. The danger with randomized planting is that this will



Perennial growth by itself can create an effect of intermingling, either through the way in which species with fine branching and spreading stems can knit together or because of seeding. Here, in August at Hummelo in the Netherlands (left), a variety of perennials in small groups are beginning to mesh together, an effect heightened



At Pensthorpe Gardens in Norfolk, England (right) two species are beginning to blend to form a meadow effect, several years after being planted in separate blocks: *Scabiosa japonica* var. *alpina* and deep pink *Dianthus carthusianorum* both have fine stems which in nature wind their way through grasses; they can also be effective at seeding themselves.

by the similar color palette of many of them. Pale *Lythrum virgatum* occupies the center, with deeper *Stachys officinalis* to the right. Behind and in front are the narrow spikes of *Liatris spicata*. Paler pink spikes belong to *Sanguisorba menziesii*. To the rear are the pale pink flowers of *Sidalcea oregana*.

not happen, and in particular that species with a strong tendency to dominance will tend to dominate everywhere, and conversely less dominant species may be forced out everywhere. Just such problems have been observed in randomized tree plantings.

Modular repetition or designed intermingling of plants is a way of getting around this problem. Piet Oudolf's work is undoubtedly a strongly designed intermingling; a mathematical analysis of his planting combinations would surely indicate an almost infinite number of potential species juxtapositions and combinations. Because of this, the long-term development of an Oudolf planting may be more akin to a naturally occurring plant community than a randomized planting. In addition, the number of species he uses is considerably greater than the 15 to 20 in most randomized mixes. Another aspect of Piet's work is that his mixtures for

larger areas are often of mingled groups of plants, of 5 to 11 individuals of each species (depending on the overall scale), so that the blend is in fact a blend of small groups.

Designers of the future will no doubt find ways of introducing greater visual richness and possibilities of future development into mixed plantings. In particular, decisions have to be made: whether to randomize, or to design mixes of small groups, or to go for small groups and individuals. Different decisions will be made for different plants and for different situations. Gardeners and designers are only at the very beginning of a whole new and immensely rich seam in planting design. Meanwhile we suggest that the thinking outlined in this book will help with this process.

ORDER AND SPONTANEITY

CREATIVE TENSIONS – BALANCING ORDER AND DISORDER

For the vast majority of garden history, gardening and landscape management have been about exerting order over nature. This was understandable in a world where nature was seen as all-powerful and not necessarily benevolent. Now, the feeling is that nature is very much in retreat in the face of an aggressive humanity – exerting order is no longer seen as always appropriate. Previous generations tended to be highly anthropocentric: to see humanity as central to the world, and therefore the imposition of what we see as beautiful or useful on to the natural world as entirely right. Now, we have come to realize that we are only a recently evolved species on an insignificant planet, and a scientific understanding of our surroundings has largely knocked us off our pedestal. While we still want our planted landscapes and gardens to conform to our ideas of beauty, we are far more prepared to accept what occurs naturally as visually pleasing. Indeed, earlier forms of imposing aesthetic order on nature are seen by some as unattractive; so highly rated now is natural as a virtue that simply describing them as unnatural implies they are undesirable.

The further away from a natural situation a planting is, the more work is required to maintain it – hence the need to clip hedges and topiary every year, and in the case of fast-growing plants, two or even three times a year. It is no surprise that so much gardening in the past was about clipping – it was, and still is, a way of showing that you were able to afford to pay people to do this work for you. It is also about control – we want to control nature and the garden is a place where we show it. The cost of maintaining plantings which emphasize human order, and a growing relaxation in our relationship with nature, are additional reasons for the greater tendency over time to appreciate the beauty of nature's apparent disorderliness.

The growth of interest in natural looking plantings has gone through several phases. Each of these has left a legacy of ideas about what is appropriate to call natural or not – sometimes these are contradictory. For example, the eighteenth-century English landscape garden intro-

duced the idea (revolutionary at the time) that the straight line was not only unnecessary but undesirable. There is still strong resistance to straight lines among many who favor natural style gardening, yet an ecologist will point out that a bird does not mind if the tree in which it makes a nest is growing in a row of others or an informal cluster, and many designers will wax lyrical about how straight lines can add a sense of order and intention to otherwise wild-looking plantings.

In Britain, the Arts and Crafts garden tradition exemplified by Lawrence Johnston's Hidcote (1907 onwards) and Harold Nicolson and Vita Sackville-West's Sissinghurst (1930 onwards) resolved a central contradiction for British gardeners, balancing their love of informal planting (mostly herbaceous perennials) with a continued hankering after formal garden features, now restricted to hedges, occasional topiary and garden structures. In the Netherlands, Mien Ruys (1904–1999) did something to achieve a modernist version of the same resolution of the conflict between formality and naturalism. For contemporary garden makers there is now a well-developed and richly nuanced range of possibilities for balancing the wildness of nature and the order of culture.

APPARENT DISORDER – THE PARADOX OF DESIGNING SPONTANEITY

The English landlords and their advisors who laid out landscapes with sinuous clumps of trees in the eighteenth century, and indeed all those who followed them, were trying to convince potential viewers, and almost certainly themselves, that what they were achieving looked natural. It is, of course, nothing of the sort, as nature would not tolerate the clear edges between woodland clumps and open grassland for very long, and would seek to clutter and blur them with seedling trees, shrubs and climbers, and eventually to reassert the wood over the grass. In addition, the clumps and their attendant lakes and fields were laid out to fit certain artistically derived ideas. Naturalism is so often a conceit.

Another naturalistic conceit, but on a different level, is the often spectacular plantings developed for garden

shows in Germany from the 1980s onwards. These were laid out for one-off events but left behind quality planting for permanent public use. The designers were trying to create plantings which looked natural. They were in no sense pretending that they were imitating nature, but instead were seeking to group plants in such a way that the results had a natural look about them, and would no doubt convince many onlookers they were looking at something very much like the real thing. In the final analysis, though, decisions about which plants to group together, in what combinations and numbers, are aesthetic ones. The creators of what have come to be known as cottage gardens have achieved similar results, but through a completely different route. Margery Fish (1888–1969) and all those who have followed her in this genre had no illusions that they were creating anything 'natural' or indeed pseudo-natural. Their model was another conceit: that they were making something akin to what humble cottagers (rural laborers and their families) would make. Both traditions end up creating an apparently artless combination of plants which have proved highly popular. The spontaneity is, however, apparent rather than real.

The truth is that while much of humanity loves nature, people do not want it untrammelled in their gardens, parks or urban spaces. The stylized natures offered by the various cultures of planting design developed during the twentieth century are the most realistic ways out of this conundrum.

STATIC AND DYNAMIC PLANTINGS – MANAGING SPONTANEITY AND CHANGE

As combinations of living things, gardens will constantly change as they age. The traditional concept of the garden tended to be static – many of the planted features were clipped into shapes that recalled the masonry of the great houses of which they formed the surroundings; trees came and went on a cycle of several human generations – while ephemeral annuals and bulbs filled rigid parterres for a few months at a time. So for many gardens there were two extremes: an almost permanent architecture and short-lived seasonal planting, both requiring high levels of maintenance. Twentieth-century gardens reduced maintenance and formality by relying more on shrubs and perennials, which allowed for

greater leeway in maintenance. They could often almost be ignored for a few years, brought back to order with a blitz of pruning, weeding and dividing, refreshed with a few new plants, and then receive a few more years of relatively unimaginative maintenance. The result was often functional but rather lackluster gardens of slightly unmanaged looking shrubs and ever-expanding clumps of a few robust perennials.

Over time, plantings based largely on perennials will age. There are two main aspects to this: some species will die out, whereas others will spread, some by simply creeping out from their original planting space, others by self-seeding. Less-skilled maintenance often only removes weeds, and does not seriously address either of these issues. In addition, slow spread can creep up on garden owners, and before they know it, the goldenrod (in old gardens derivatives of *Solidago canadensis*) and Japanese anemones (*Anemone x hybrida*) have taken over. The result is chiefly a loss of diversity. The issues of vegetative spread, self-seeding and plant death are looked at later on.

Perennials, and to some extent shrubs, have their own life cycles of growth and replacement and spontaneously self-seed between different parts of the garden. This dynamism has been played with cautiously for much of the twentieth century, but now with a better understanding of plant ecology gardeners and designers are in a stronger position to work with spontaneity rather than resist it as 'disorder' in the way our gardening ancestors did. A key problem remains, though, the tendency to lose species over time and so lose visual diversity.

How much is lost over time from a planting depends very much on what was planted originally, in particular whether long-lived and robust species were chosen or not. During the early years of the twentieth century, the selection of perennials was dominated by short-lived species or those which were not very persistent. In the latter part of the century these were replaced with more robust and naturally long-lived or persistent species, often less intensively hybridized. At the time of writing, older parts of the Oudolf garden at Hummelo have hardly had any replanting done since they were laid out 20–25 years ago.

How long do perennial plantings last? And – what should be done when they need restoration? Two types

It is not entirely true that species diversity in plantings declines over time. Sometimes, attractive and not-too-aggressive wild species may appear and add themselves to borders. One such is the early spring flowering *Cardamine pratensis* (shown here in close-up), especially on damp soils.

Upright *Verbascum leichtlinii* is one of several species of a large genus of plants in a family (*Scrophulariaceae*) where a great many are either biennials or short-lived perennials. Survival is through scattering their plentiful seed to the wind from sturdy seedheads. Their advantage to gardeners is the spontaneity and naturalness of the way that they move themselves around the garden. Here, at Hummelo, in early June they flower alongside *Knautia macedonica* (deep red, foreground, a short-lived perennial). The combination of clipped yew hedge and self-seeding plants makes for a great expression of creative tension.



of management can be envisaged: a low-key, relatively passive one, and a more active one, but which works very much with the naturally dynamic nature of the planting. A passive approach involves:

- weeding
- removal of the most vigorously self-seeding species
- annual cutting back of dead growth
- bringing the dominant plants back into the right proportion.

Even the best planned perennial-based planting will experience a decline in species diversity – highly persistent and the most vigorously self-seeding species will be the survivors. Cutting back annually and leaving the shredded debris as mulch or taking it away, composting

it and returning it as mulch will keep nutrients recycled. Even if material is taken away and not returned, causing nutrients to decline slowly, growth will often continue to be very good for many years on many soils, as long-lived, persistent perennials do not have a great hunger for nutrients.

A more active involvement with the planting recognizes natural processes of death and rebirth: ecological processes, in other words. James Hitchmough and Nigel Dunnett at the University of Sheffield's innovative Department of Landscape refer to this as a 'dynamic planting'. Change and spontaneity are embraced. However, the planting will not look after itself. Engaging with a dynamic planting involves controlling and editing the





In the new perennial meadow in the Oudolf family garden at Hummelo in the Netherlands, perennials arise out of wild grasses and spontaneous flora, such as the wild chamomile (white flowers, foreground). Other species here are *Agastache nepetoides* (seedheads, left), *Aster 'Little Carlow'* (center), *Aster novae-angliae 'Violetta'* (right), and pink-red *Eupatorium maculatum 'Riesenschirm'* in the foreground, right.

Avenues are a traditional way of setting out landscape plants. A row of trees makes an effective backdrop to the apparent chaos of a late autumn border of grasses and perennials at Westerkade, Rotterdam.



outcomes of natural processes, steering the planting in a way which maintains or even enhances its appeal. We will look in more detail at how this can be done later. Key to managing a dynamic planting is understanding that perennials have a variety of life cycles.

However, even with the most skilled management, a point is reached when some sort of restoration becomes necessary. This is very difficult to predict, both because of the complexity of plant-environment interactions and because the perception of the planting is inevitably subjective. Ten years might be a good time if expectations and demands are high. What happens now? 'Restoring' is still favored by many, but this assumes a

turning back: recreating how it was when first made. In the (for the sake of argument) ten years since the planting was made, the following will have happened:

- It will have become obvious which species are problematic.
- New species and cultivars will become commercially available.
- Ideas about planting design will have changed.

Exact restoration will almost certainly be inappropriate, and will often seem undesirable. Renovation might be a better word. New cultivars and ideas need to be taken on board, so that the original plants live on alongside newcomers, and the planting reflects the passage of time.

THE CONTEXT FOR PLANTING

THE HUMAN AND DESIGN CONTEXT

Planting has to please people – as has been said before, people are part of the ecology too. The point has been made by others that in order for natural environments to be valued by humans they have to be liked – simply functional plantings which satisfy technical criteria for sustainability or biodiversity but do not satisfy human users are in the long run doomed, because nobody will care for them enough to campaign for them when they are threatened by other potential users on this overcrowded planet or simply through lack of care.

The role of the gardener or designer is clear, and arguably more important than ever: planting which serves a purpose has to look good too. As practitioners use plant combinations more and more to perform environmental services, it will become increasingly important that they look intentional and attractive.

Tidiness and presence are important issues, although often read very subjectively. In corporate or memorial plantings, even occasional seasonal scruffiness is un-

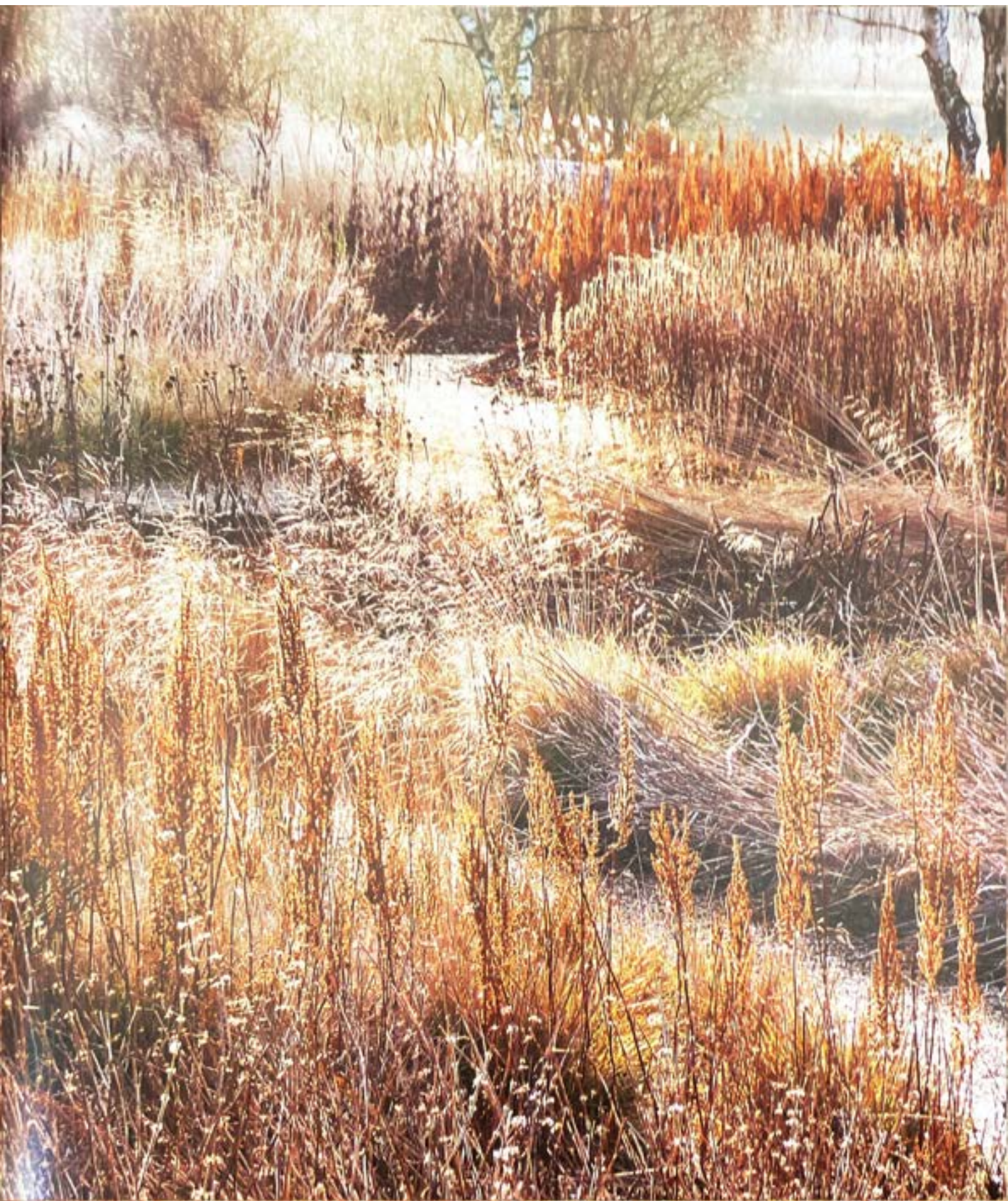
acceptable. If there is no guarantee of labor to keep up standards, planting will have to be limited to species which can be kept acceptably tidy, although it is possible to include species which look unattractive post-flowering in large areas of mixed planting where there is enough structure to maintain the look of the whole. There is more scope for using less tidy-looking plants in more relaxed surroundings. This is an important point, as some of the most popular, easily propagated and resilient perennials do look bad after flowering. Keeping such plants to no more than an absolute maximum of 30 percent is a good rule, but even so, for some situations where such plants may grow very well and flower for long periods, gardeners and designers may feel they want to use more. Some such plants, certain *Geranium* species in particular being a good example, are also very long-lived and effective ground cover and can be tempting to use in large quantities in low-maintenance situations. The addition of just a few distinctive structure plants, or others with a different (usually later) flowering time, can do much to lift such a planting beyond a



◀ and ▶▶ (pages 42–43)

Bright colors are popular, and can be entirely compatible with naturalistic or functional planting. This is *Astilbe chinensis* var. *taquetii* 'Purpurlanze' in July, a robust and long-lived perennial eminently suitable for drainage swales and other sustainable drainage plantings. Its winter seedheads (overleaf) have great value.









◀◀ (pages 44–45)

During 2011 the Serpentine Gallery, a venue for contemporary art in London, hosted a Piet Oudolf planting in a courtyard. For the majority of the species used, the location would not have been suitable in the long term, but a summer-long show garden-type display was perfectly feasible. As with flower-show gardens, plants were used at higher densities than would be normal for a permanent planting, but one advantage (as show garden designers know only too well) is that the visual experience is more intense than in the real garden – this is, of course, useful in a venue such as this where onlookers are at very close quarters to the plants.

►
Temporary plantings do not have to be composed of garish annuals. Here, at the 2010 Venice Biennale, is a rarity – an Oudolf annual composition of *Hordeum jubatum* (wild barley) and dahlias. It playfully evokes the spontaneous plant communities of waste ground.



flat monoculture. Other possibilities are partial concealment by using them in drifts around more strongly structural plants or placing them where they can be enjoyed when they perform but do not detract from key vistas. These plants can often be managed more intensively for a higher quality look in smaller spaces, and of course in private gardens, where there is more possibility of managing them through seasonal cutting back.

CONTROL OVER NATURE? – FORMALITY AND MASS PLANTING

Formal planting – the use of geometry or clipping to create unnatural forms – is widely seen as traditional. There are, however, many possibilities for a more contemporary approach. Novel forms of formality (such as non-symmetric clipped woody plants or blocks of ornamental grasses) are nowadays more likely to be seen as appropriate in larger and grander contexts than traditional, clipped and symmetric formality. Larger scale

settings or those which need to look imposing require simple graphic qualities, which can be expressed through some kind of monocultural planting. Traditional materials like yew or beech can be given new life with modernist-inspired geometries. Ornamental grasses are the other great material for monocultural blocks, because of their relatively long season of interest, adaptability, longevity (for most species) and simplicity of form. An added advantage is that they take less maintenance than clipped foliage. Their disadvantage, vis-à-vis clipped woody plants, is that they do not have the same level of presence all through the year.

One of the great uses of any kind of mass planting like this is that it can contrast so effectively with highly complex and diverse plantings of perennials, giving architecture and backbone to what would otherwise be too visually soft-textured for some environments. It is worth considering here the global success of the English Arts and Crafts garden style. Fundamental to it is a balance between order and apparent disorder. Rather than



Molinia caerulea "Dauerstrahl" in blocks. This is a native plant of acid soils across northern Europe, so its presence here in the Netherlands can be described as a novel use of a formal planting principle with a plant which actually relates to its landscape, much as previous generations might have had clipped hedges of native trees. This kind of simple, calm, ordered planting can be a good point of relaxation after an exuberant perennial-filled border – like the sorbet course in a gourmet meal.

relying on the old vocabulary of clipped woody plants, contemporary gardeners and designers should perhaps be looking at grass blocks to provide the structural part of the equation.

In corporate or monumental landscapes, simple blocks of woody plants with grasses may be all that is needed to create the atmosphere of calm, control and order. Where the general public is concerned, however, these words are more likely to have negative connotations, as many people have an expectation of planting providing life, vitality and color. Where the scale is large, as in public parks, it can be difficult for colorful and diverse planting to actually make much of an impact – or to rephrase this, there is rarely enough funding for plantings that really make an impact. The creators of randomized planting mixes and seed-sown perennial mixes (such as prairies) claim that their combinations can be maintained extensively – all plants attended to at once, mainly through mowing. This may well be the case, at least in some climate zones, but the concept is

still too new for a long-term picture of their maintenance needs to be clearly understood. If maintenance can be kept low enough, then these plantings do genuinely offer an option for large-scale situations.

NATURE OVER CONTROL – MAKING WILD-LOOKING PLANTINGS

Those who wish the plantings they make or commission to look tidy and ordered – ebullient nature controlled and organized for the human eye – stand in the majority tradition of garden history. The more recent tradition, which I see Piet Oudolf and myself as part of, is about the opposite – how to create the sensation of nature, often in urban or suburban areas. As we know all too well, this does not mean letting things go by allowing natural processes to take over. When people say they want some nature, what they usually mean is a particular vision of nature, one that looks nice, fitting in to a distinctly human-centered idea of what nature is or

Graphic simplicity is given an added boost through autumn color. Here, *Rhus typhina*, a suckering shrub which can be restricted through coppicing, grows alongside *Calamagrostis* 'Karl Foerster', a grass which is extremely useful for minimalist or 'neo-formal' plantings because of its long season (early summer to late winter) and its weatherproof, upright habit.







▲ A distinct feature of the garden at Hummelo is the big clump of the grass *Miscanthus sinensis* 'Malepartus' in a slightly raised position, seen here through a haze of the seedheads of another grass, *Molinia* 'Transparent'. The miscanthus block acts

like a weight, a center of gravity for the garden as a whole, especially since it is more or less midway between two distinct areas of the garden, both dominated by looser perennial borders.



▲ Wild and woolly, the High Line is dominated from late summer on by the seedheads of grasses – anything tidier here would look out of place next to the still scruffy areas of the urban jungle. The flash of orange-red is *Iris fulva*.

should look like. Biodiversity is important too (but not mosquitoes or snakes, of course!), and often locally native plants. 'Nature Lite' in other words. We must not be too cynical – not too long ago any idea of 'nature in the city' would have been anathema or at least incomprehensible to most people. The task for the gardener or designer is to create an enhanced nature (a term coined by Nigel Dunnett and James Hitchmough), one that supports an acceptable level of biodiversity and looks just a little bit wild. How can this be achieved?

- Use plant species which are recognizably native locally – even if in some cases they may be commercially selected cultivars.
- Use a far denser and more intermingled style of vegetation than has been conventional up to now – in particular, layered vegetation, open habitats dominated by grasses blended with perennials, woodland underplanting with a mix of species.
- Use plants which recognizably belong to a particular habitat – for example, grasses in expansive open spaces,

ferns and evergreen ground-cover species on woodland floor. Note that these do not have to be locally native species.

- Create plant combinations which evoke wild habitats – such as water bodies edged with lush reedy and leafy vegetation, small shrubs and climbers occupying a mixed zone between woodland planting and open habitat.
- Allow an element of spontaneity – self-seeding and a sense that it is partly the plants that are in control.
- Have a wild or semi-natural background – the viewer can then be persuaded that there is a seamless link between the planting and the existing natural vegetation.

In a public setting much of the above is about getting people to read something subconsciously, and the people concerned will for the most part have very little real knowledge or experience of natural plant communities. What matters to them, what will convince them and enable them to read the designed planting as some kind of 'nature', is that it evokes the country park or nature re-

serve or state forest or wherever it is that urban and suburban people in a particular place experience nature. The New York High Line is a good recent example of this kind of planting. The world's oldest established and most extensive examples are the parks of Amstelveen in the Netherlands, dating back to the 1930s, where the planting is almost entirely native but is highly managed for a relatively neat appearance.

For homeowners who are only trying to convince themselves and their family and guests, the hardest task is probably to convince themselves that they have built and planted a microcosm of nature. Gardeners or others who are interested in nature and know their wildflowers may be hard to convince. Most who do succeed often mix natives with garden plants, and have cultivated the art of knowing how much to let go.

SIGNATURE PLANTING – DEVELOPING AN IDENTITY

The idea of signature is a strong one in the art world, the distinctive stamp of an artist or maker. Good garden designers have all developed a very strong signature, so that the knowledgeable could probably recognize one of their gardens if they were parachuted into one they had not seen before. Gardens too can develop a signature in the minds of those who visit them, because of some distinctive feature, or the use of a strong theme plant which is scattered throughout the garden.

While private gardeners who develop a theme for their garden which is endlessly admired and photographed by all who visit can sit back and pat themselves on the back, the problem for designers who develop a successful signature for a particular garden is that it becomes very difficult to reproduce elsewhere. The other problem is that everyone else copies you, and usually less successfully, perhaps even turning an innovation into a cliché.

Signature plantings are about making something that is site-specific and gives the garden or landscape a distinct personality which helps to make it memorable. This is clearly important for public landscapes, but home gardeners can try it too. One plant spread throughout a garden can make a real impact in the here-and-now but also in the memory of visitors. I have an immensely tall and narrow clone of *Eupatorium fistulo-*

sum, which grows to 3.4 meters high every year; visitors to the garden in the late summer to winter period inevitably remark on it, and even American visitors are impressed (it is a US native). Over the years it has become the most commented on plant in my garden, and I am sure the most remembered.

Signature plantings may be derived from several sources:

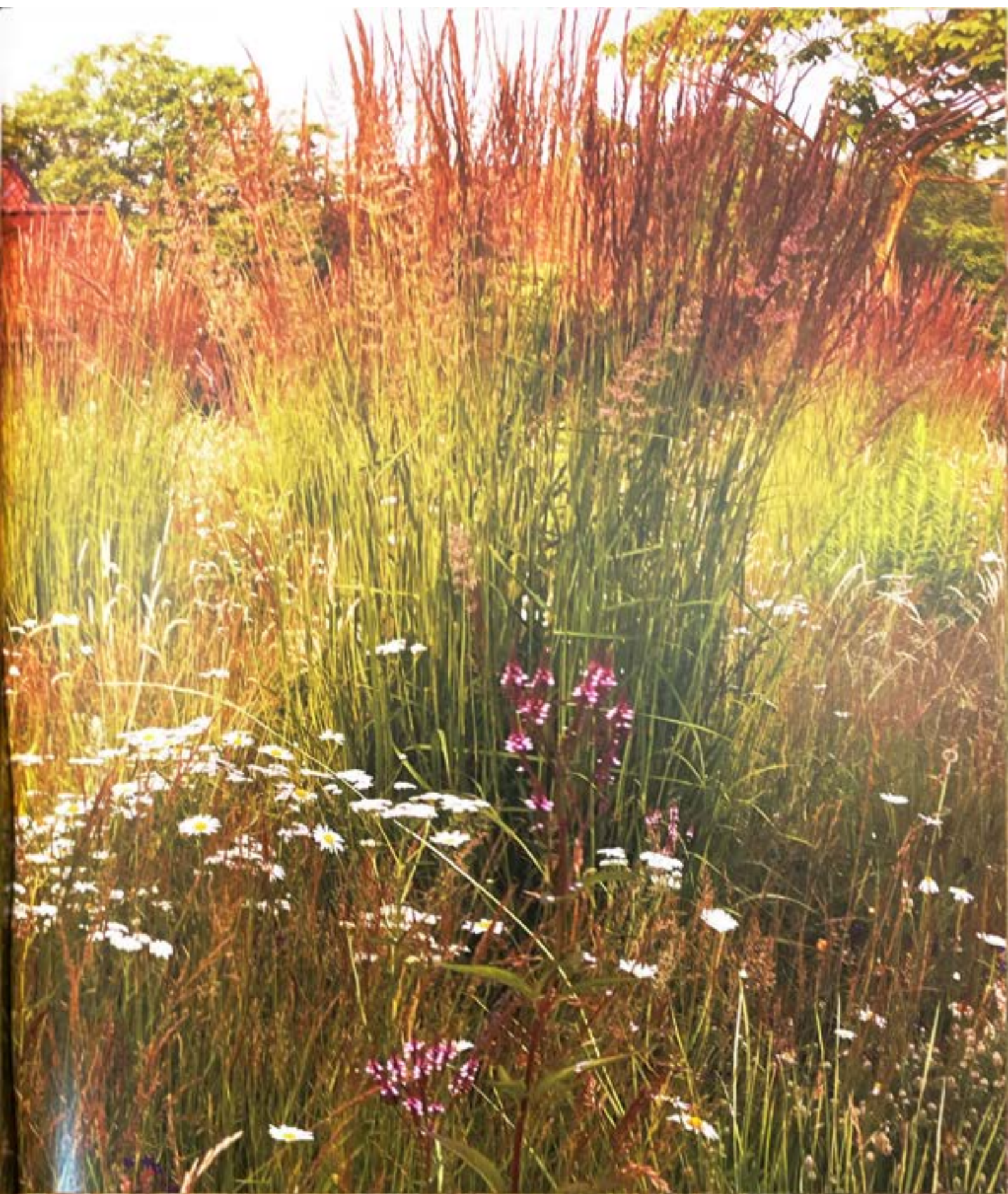
- Historical. Certain plants may be associated with a particular site or period associated with the site, or simply be historical plants in gardens, although usually these are woody plants, such as trees or species used in traditional hedging or topiary; *Liriope* and *Ophiopogon* species in the Far East are a herbaceous example. Radical new uses of these traditional (even clichéd) plants can make a strong impression.
- Locally native. There may be a desire to use certain species or plant communities which are part of the local scenery, so celebrating what is special to the region.
- Ecologically appropriate. Using quantities of a distinctive species which can be clearly linked to the site ecology. Examples include swathes of the very distinctive grass species which dominate many stressful environments, such as *Molinia caerulea* (poor acidic soils) or *Sporobolus heterolepis* (extreme Continental climates).
- Daring to innovate. New plants or familiar plants used in a new way. A new umbellifer in the show garden at the Chelsea Flower Show in London in 2010 designed by leading British designer Tom Stuart-Smith got everyone talking (*Cenolophium denudatum*). It was not in fact very different to many other members of the family currently used in gardens; the fact that it ran through the show garden and was used with great confidence made it a perfect signature plant.

The following Piet Oudolf projects exemplify what have become recognized as strong signatures in the garden and landscape press, or otherwise stand out for me as signature plantings.

Bury Court, Hampshire, UK, 1996

This stylized meadow originally used *Deschampsia cespitosa* interspersed with a small number of contrasting perennials. Widely agreed to be a good idea at the time, it was copied by several other designers, generally unsuccessfully. One problem is that the species can die prematurely on fertile soils, and some cultivars are





◀◀ (pages 52–53)

Piet's use of a grass matrix in plantings started with what became a well-known new feature, the *Deschampsia* meadow at Bury Court (1996). The latest manifestation is a new planting on the former nursery site at Hummelo. Created through both planting and sowing, the species mix will change from year to year. Tall grass *Calamagrostis* 'Karl Foerster' and the scarlet cultivar of *Helenium* will be more or less permanent, but many other shorter-lived species, such as the two color forms of *Verbena hastata* and Dutch native daisy *Leucanthemum vulgare*, will be dependent on self-sowing for their continued existence here.

▶

The double border at the RHS Garden, Wisley (2001), where bands of intermingled perennials create interesting combinations at all scales of observation, from near to far. In mid- to late summer *Perovskia atriplicifolia* dominates (right above), counterposed with pink *Gaura lindheimeri* 'Siskiyou Pink' and the seedheads of *Phlox tuberosa* 'Amazone', while in another band (right below) gray-white *Eryngium yuccifolium* contrasts with scarlet *Helenium* 'Rubinzwerg' and *Echinacea purpurea*; some seedheads of *Allium hollandicum* survive from an earlier phase of flower.

▶



susceptible to fungal diseases and can suffer extensive dieback. *Deschampsia* has now been replaced by cultivars of *Molinia caerulea*, which, although also a plant of infertile soils, is longer lived across a wider range of habitats. This concept makes an impact in design terms and makes sense in management, and it can be emulated by using other grass species and therefore work in many different climate zones.

Conclusion – a signature might be a very good concept, but needs to be adaptable to be effective long term.

Royal Horticultural Society Garden, Wisley, Surrey, UK, 2001

Thirty-three bands of intermingled perennials and grasses, each of equal size, line either side of a straight grass walkway. Looking very formal and rigid on paper, it is anything but in practice. A good example of a simple design concept which underlies an entire scheme, but largely succeeds in hiding itself so that perhaps comparatively few people have really understood it.

Conclusion – strong signatures are sometimes almost invisible, operating on the level of the subconscious.

Dream Park, Enköping, Sweden, 1996 and 2003

The Salvia River is a band of three violet-blue cultivars of European origin. The public loved it – it had impact and the photogenic ‘wow’ factor that commissioners of such projects always want. Something so simple and dramatic is very hard to repeat, but Piet did so in the Lurie Garden in Chicago; as he describes it, ‘It was one of the few times that I really copied myself, it had to do with the graphical quality of the concept, it had to be seen from the surrounding skyscrapers.’ No one else has yet copied it or tried to achieve it using other species. One of the problems in doing so is that most colorful perennials in a low to medium height range have poor structure post-flowering (the salvias just look a bit dull, and can be cut back to repeat flower later).

Conclusion – there is always a danger that very strong signatures will be copied, but it is a risk worth taking.

County Cork garden, Ireland, 2006

Stipa gigantea is a tall and expansive but at the same time very light and transparent grass. Piet used it in gardens back in the 1980s and early 1990s, but when everyone

started using it, he gave up on it. Here, however, just a few are used to spectacular effect; at a latitude of 51 degrees north, early morning and late afternoon sunlight striking this grass illuminates it to dramatic effect in this open site.

Conclusion – clichés can be challenged.

Oudolf garden, Hummelo, the Netherlands, 1982 onwards

The multiple curtains of yew at the rear of the main garden area at Piet and Anja’s own garden became, in that much overused word, iconic. In the end, though, they succumbed to the forces of nature in the form of flooding which led to phytophthora disease and dieback, so early in 2011 they were removed and fed dramatically into a chipper. Piet phlegmatically accepted that it was in danger of becoming a cliché.

Conclusion – icons sometimes need iconoclasts.

The High Line, New York City, USA, 2009 onwards

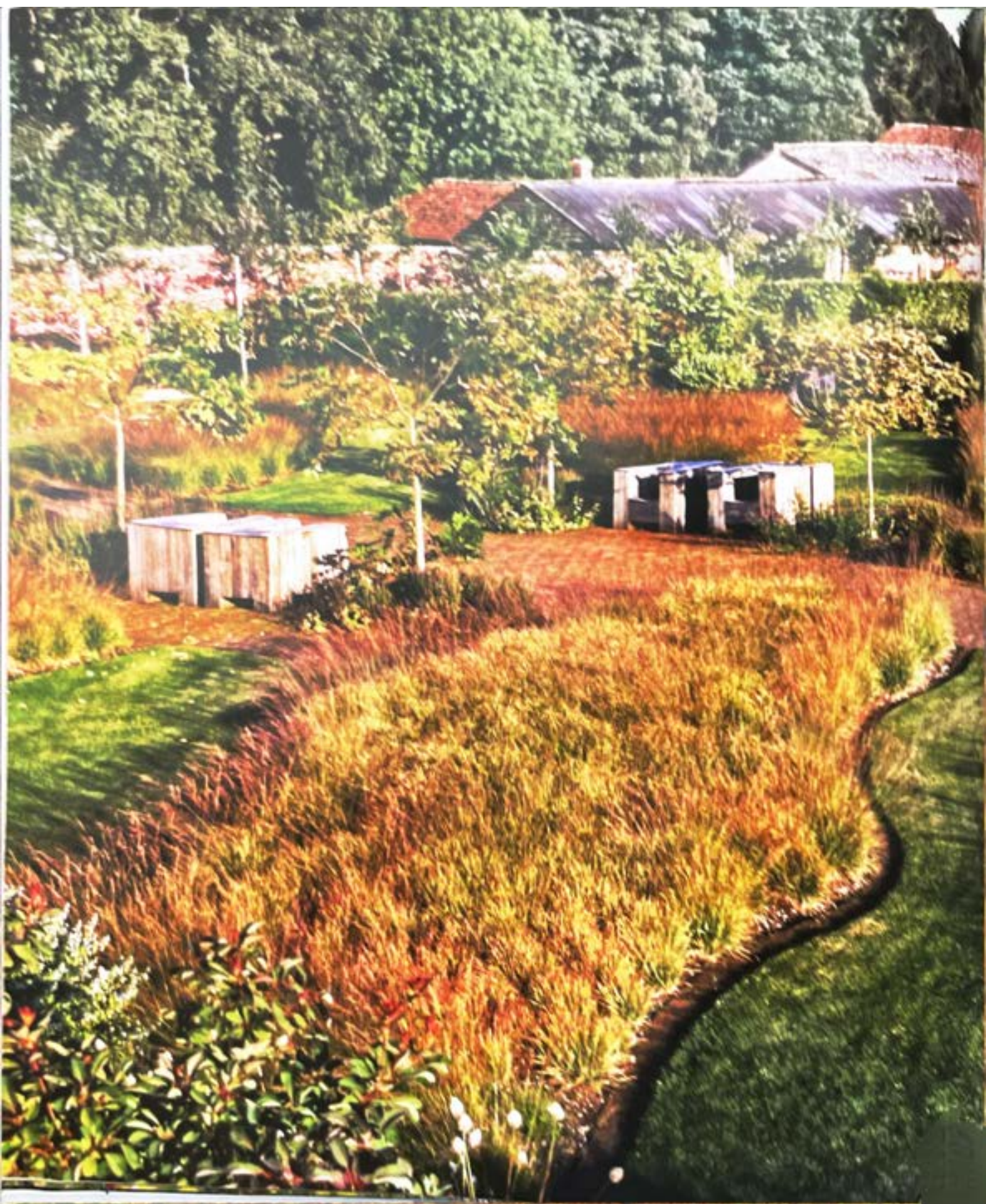
It is perhaps the grasses on the High Line which make it so distinctive. Originally (that is, as an abandoned freight rail line), grasses were part of a rich species mix, so using them was part of the design intent to keep a strong sense of the original spontaneous planting. They certainly make the High Line the most naturalistic Oudolf planting so far, and add hugely to its character. Part of the appeal of the High Line is that it is so successfully brings nature into the city – the grasses are crucial for this.

Conclusion – signature plants can be very useful for evoking atmosphere, so long as the vast majority of viewers can read them.

Scampston Hall, Yorkshire, UK, 1999

A relatively low-growing combination of plants makes the most of a poor soil, in a place which can get cold winters and is on the drier side of England. It is the kind of planting, like Beth Chatto’s in Essex, which fits plants to habitat. Colorful and rich in textures, it makes the best of its lean look. A dramatic use of waves of a *Molinia* grass variety is another play with a simple modernist-formal feature.

Conclusion – poor soil and less than perfect conditions can be reflected in planting in a positive way which helps to create a sense of identity.





♦♦ (pages 56–57)

Bands of *Molinia caerulea* 'Poul Petersen' at Scampston Hall in Yorkshire, northern England, were designed in 1999 but retain their freshness as a design feature, powerfully emphasizing the potential of grasses as material for contemporary formal structure.

►

The rear hedges of yew (*Taxus baccata*) at Hummelo (above right) became one of its best-known features. A classical material and idiom, their asymmetric cut and modernist flair deservedly made them a contemporary garden design icon. After their removal (below right), the garden is defined by a mix of hedging – a clipped but informal version of a traditional mixed farm hedge (on the left here) and more formally clipped beech hedging at the rear (only just visible).

►





Part of the Salvia River in Chicago's Lurie Garden, where a mix of early summer flowering *Salvia* cultivars are used to dramatic effect: *S. x sylvestris* 'Mainacht' ('May Night'), 'Blauhügel' ('Blue Hill') and 'Rügen', and *S. nemorosa* 'Wesuwe'. The grasses are mostly *Sporobolus heterolepis*.

►► (pages 60–61)

Stipa gigantea needs to be positioned right to be fully appreciated, even more than other grasses. Arguably, it is more successful at higher latitudes where side or back lighting illuminates it more effectively – this garden is in western Ireland. The deep blue to the left is *Agastache* 'Blue Fortune'.

FROM NEAR TO FAR – COMPLEXITY AND SCALE

Intermingled planting styles offer complexity and diversity. But how easily is this read by the viewer? Is there not a danger that it just looks like a chaotic mush? Much must depend on the plants used in it. There is a clear need for structural plants, as relying on color alone risks the planting dissolving into amorphousness once the flowers die. More fundamentally, though, what does the eye see at different scales? It is important to stress here that there is a difference between random complexity and designed complexity.

On a large scale, there is the phenomenon of seeing the wood not the trees: viewing a mixed planting as a vegetation, as a community, not the individual plants in it – one reads only the patterns. What in fact is a highly complex mixture becomes resolved into a unified whole that is greater than the sum of the parts. Within this whole, individual placing does not matter, and randomization works as well as anything else. If it can be seen to work in the long term, this tendency suggests that the mixed planting style has a great future for use in large

spaces. In smaller or more intimate spaces, the very randomness of the mix lets it down. As the scale gets reduced, so the importance of limited key spaces within sight becomes greater. What occupies those spaces is too important to be left to chance. Blending and intermingling varieties is possible, but they need to be much more precisely placed. As far as individual plant placement is concerned, control over exact location becomes more important the smaller the planting. At this level, layering becomes of greater importance than at larger scales. If there are plants at different heights within the planting with varying habits – upright, clump-forming, ground-hugging, sprawling – plenty of scope exists for creating rich visual interest.

The distance from which we view a planting makes all the difference in our perception of its scale. Have you ever seen a wildflower meadow from the air? No. Because at that height the flowers blur into the grass and it becomes a green whole. In the garden, some plants sink into insignificance at certain distances, not just because they are small, but because they lack outstanding color or form. Others can make an impact at a distance, but only if grouped, whereas some stand out as individuals.





PLANTING AND SUSTAINABILITY

Sustainability has become a key concept but also a much-abused one, partly because it has become heavily politicized on the one hand and reduced almost to a cliché on the other. It is generally understood as meaning reducing inputs of non-renewable resources and harmful outputs. Like so much else, it is best seen as part of a gradient. Passive sustainability can be understood as not doing actual harm through causing pollution or CO₂ emissions and over-exploitation of resources.

Active sustainability can be thought of as to do with not just minimizing harm to the wider environment, but actively managing and improving it through some of the methods discussed in the Introduction: green roofs, rain gardens, biofiltration. These fields are all quite specialist, but can use the kind of diverse herbaceous vegetation highlighted here very effectively. The input of gardeners and designers into such active systems is to provide the aesthetic dimension. Plant selection has fundamentally to serve certain functions, based on a knowledge of plant physiology, best understood by ap-

propriate specialists. However, leaving plant selection entirely with technicians rarely achieves the most aesthetically pleasing results, and gardeners and designers have a role to play in working with plant lists provided by the specialists.

BIODIVERSITY AND THE NEEDS OF NATURE

Whereas once the needs of wildlife were not recognized, contemporary expectations are that planting should support some part of the web of nature. The good news is that providing for biodiversity is not difficult – in fact, many gardeners were doing this long before the word was even invented. To summarize some of the scientific work on the issue, the most important aspects of planting design for animal diversity are a range of habitats, such as is provided by a combination of trees, shrubs, perennials and ground-cover vegetation, and – crucially – connections between habitats.

Wilderness in the city – this is an area of the High Line in October. A variety of seedheads provides food for birds and small mammals. A high percentage of native species will help support those specialist invertebrates which will only eat specific plants. The brown grass on the right is *Chamaenanthium latifolium*, a relatively shade-tolerant North American species.



Promoting diverse perennial vegetation which stands over a long season is a good start, with woody plants and tree cover also having an important part to play. The use of regionally native plants may help but is not absolutely vital.

Nature thrives on diversity, and so it should come as no surprise that the intermingled planting approach has enormous potential for supporting biodiversity. It is not that conventional block planting is necessarily deleterious, but the possibilities for improving resources for biodiversity are greatly improved with more diverse intermingled plantings.

SECOND GUESSING THE FUTURE - CLIMATE CHANGE AND DIVERSITY

We live at a time when the impact of possible climate change hangs over us like a threatening thunder cloud. The garden and landscape world has, with agriculture, a part to play not only in reacting to climate change but also in its mitigation.

Climate change is often mistakenly described as 'global warming'. A two-decade run of mild winters in north-west Europe led to much complacent planting of warmer climate species during this period: there was almost a sense of some gardeners even relishing a new warmer climate and the ability to pick their own olives and pomegranates. At the time of writing this region has experienced a run of cold winters, and climate experts now say that climate change may lead to colder winters here. The gardens of the English Midlands are full of dead *Eucalyptus*, *Cordyline* and *Phormium*. The one certainty about climate change seems to be greater uncertainty - the prospect of more extreme weather. We need resilient plants.

Fortunately, there are a great many resilient plants, and we already garden with quite a few. There is more scope for not only making new introductions from the wild but introducing a wider genepool of the species already in cultivation. Many of the plants relied on are descended from one single introduction, and may be quite unrepresentative of the species as a whole. We could be missing out on forms which are not only distinct in visual terms but able to cope with different or more extreme environmental stresses. *Astrantia major* is a species common in central Europe; new seed collections

and therefore new genetic material for cultivation can be made by driving into a layby in Austria and leaning out of the car window - not surprisingly, there is a considerable range of color and vigor of the species in cultivation. *Phlomis russeliana* is another perennial which in cultivation shows almost no variation whatever, despite being an extremely useful plant, particularly for plantings requiring very low levels of maintenance. Introducing new genetic material from the wild would require more planning and more effort, and traveling further away from the localities where the plant would most likely be used. It may well come about that up-and-coming growers in *Phlomis*'s native Turkey will eventually make these collections. We need to create plantings resilient to changing weather patterns by making far more effective use of nature's bounty of diversity.

There is one further aspect to the importance of genetic diversity among cultivated plants - diversity can increase resilience in plant populations which are dynamic and self-replicating through self-seeding. Currently, professional planting design assumes that the plants used will be permanent. Private gardeners recognize that the seeding of short-lived perennials or biennials can play an important part in the appeal of their plantings. The acceptance of a dynamic element in larger or public plantings implies self-sowing and a natural process of plant replacement. This is particularly likely to be seen as desirable in randomly mixed plantings. A planting where self-seeding is happening involves a reshuffling of genes and so the appearance of seedlings ever so slightly different to their parents. Some of the seedlings will be better able to cope with the extremes of weather than their parents, and so, through being dynamic, the planting will be able to adapt over time in the way that a natural plant community does. In theory! In practice, most elements will be so long-lived that any change will happen on a time scale far too long to be meaningful. However, this process of adaptation over time (which, incidentally, beautifully illustrates Darwin's theory of evolution) can make a real impact for shorter lived species or those which readily self-seed.

A good example of the potential for Darwinian natural selection to make a meaningful impact on gardens is provided by the South African genus *Dierama*, a member of the Iris family (*Iridaceae*) of great beauty, which commonly thrives and enthusiastically self-sows in gardens





Preparing for climate change should arguably be about selecting plants which survive multiple stresses. A good reference habitat is the steppe of eastern Europe and central Asia with its cold winters and dry summers – analogous to short grass prairie and sagebrush country. An advantage of many steppe and other drought-tolerant species is attractive silver-gray foliage, such as *Artemisia pontica* here. The yellow is *Euphorbia seguieriana* subsp. *nicaeana*. The yellow-green grass foliage is *Sesleria autumnalis*, the pale grass spikes *Melica ciliata*, and the very long silvery grass heads are a form of *Stipa pulcherrima* – this magical effect is unfortunately short-lived. This is early summer.

on north-west Europe's Atlantic rim. The plants are variably hardy, but naturally grow in dry-winter climates, adding to the complexity of their response to cold weather. A run of cold winters can have a real winnowing effect on a plant population, with survivors flourishing and reproducing, non-survivors (and their genes) making their way to the compost heap. The result within a garden can be a reliably hardy and resilient population of a beautiful and elegant plant.

Much planting design practice to date has relied on genetically identical plants – cultivars. Cultivars are favored by gardeners and designers because of their consistency, yet they have disadvantages. That clones or extensively grown varieties with narrow genepools are susceptible to the spread of disease is well established in agriculture, especially after the USA nearly lost most of its corn crop in 1968. In addition, some cultivars will not self-seed because of a natural tendency for many perennials to 'out-breed': they will not self-seed unless there is genetic difference between the prospective parents. Even if they do produce viable seed, the genepool of a population of seedlings from a cultivar will be restricted. A more 'natural' varied population will have within it a much wider range of genes, with all this implies for a greater adaptability. Again, *Dierama* offers a good example, as several species are in cultivation and they crossbreed very easily; the result is a range of genetic diversity from which cold winters can select a strain which will survive long term.

THINKING THROUGH RESOURCES – SUSTAINABILITY QUESTIONS

Another major issue raised by sustainability is that of resource use – essentially a complex of issues comprising the exploitation of non-renewable resources, including those for energy production, and the generation of CO₂ and other pollutants through transport and other energy-consuming activities connected with garden and landscape construction and maintenance. Perennial planting almost by definition looks to the long term, and is therefore less resource hungry than traditional seasonal bedding. Little auditing has been carried out, but it is likely that the continual use of mowing machinery for keeping turf grass short involves greater resource consumption and CO₂ output than the maintenance of

perennial plantings. The creation of perennial plantings can involve considerable resources, and this is the main area where we all need to look closely at sustainability issues. Consideration of these issues needs to be carried out objectively and on the basis of evidence; sustainability and other environmental issues are notoriously political, and much decision-making by both private individuals and institutions is carried out in an atmosphere clouded by emotive and ideological considerations – the British debate over the use of peat in potting composts is a good example of this.

The horticulture industry used to be notably insensitive to resource use. Plastic pots and other containers that cannot be recycled, energy for heating and lighting propagation facilities, and compost all gobble up resources and require physical materials and energy for their production and crucially their transport. Over recent years great strides have been made: reusable or recyclable containers, the widespread use of composted green waste for potting and soil improvement and of slow-release fertilizers which reduce the wastage of nutrients, and much more. However, there is one tendency which still needs questioning – the size of plants in new projects. It is accepted that small plants usually establish better and more quickly than large ones. Yet, so many new planting schemes, both private and public, involve the purchase of container-grown plants far larger than would have been traditionally used. Moreover, these plants often travel long distances from nursery to site. So, two issues have come together which massively increase the embodied energy of many commercially grown plants. Every perennial in a two-liter pot as opposed to a half-liter one weighs four times as much and takes up about four times the space in a delivery truck; delivery from a nursery 400 kilometers away uses four times as much fuel as from one 100 kilometers away. Do the math!

The case for using small plants is easily made for perennials, which tend to grow quickly, but is less easily made for woody plants. Many design professionals are under pressure from clients for quick results (those spending public money are often worse than private), and it can be difficult to resist. What can clinch the argument is pointing out that small woody plants often catch up with larger ones in a few years, such is their superior ability to adapt and make the best of new



On the High Line, regionally native plants make an impact. Fine, blue-gray grass at the centre *Schizachyrium scoparium* 'The Blues' with *Bouteloua curtipendula* (the grass with fine, one-sided seedheads), yellow *Rudbeckia subtomentosa* and in the background *Vernonia noveboracensis*. All of these are now well established in cultivation.



The days of automatically cutting down dead perennial growth in the autumn are long gone. Most gardeners and those involved in plant management are now familiar with the idea that seedheads feed birds and may harbor a variety of invertebrates too. The idea that the city can be a habitat is widely accepted in the industrialized world. It is also worth pointing out the value of misty weather adding a sense of drama to combinations of tall perennials, such as this *Angelica gigas* on the Westerkade in Rotterdam.

Early autumn at Hummelo with the lush perennials and grasses which create so much of the distinctive character of this time of year. Grass *Molinia caerulea* 'Transparent' with pink *Eupatorium maculatum* 'Riesenschirm', the tiny red heads of *Sanguisorba* 'Bury Court', white *Panicum amplexicaule* 'Alba' and yellow *Solidago x luteus* 'Lemore'.









◀◀ (pages 70–71)

Regionally native plants bring nature and a hint of the wilderness to the High Line in early autumn. *Rhus typhina* is already changing color, as the flower-heads of *Eupatorium hyssopifolium* repeat down the line. *Aster oblongifolius* 'Raydon's Favorite' is just visible – an example of an increasing trend in naming cultivars of native plants.

Plants in designed landscapes may even play a role in conservation. The graphic qualities of the royal fern (*Osmunda regalis*) are used here in a private garden (Dyffryn Farnant, Pembrokeshire, Wales), creating visual weight amongst a mass of finer foliage. This species is now very rare in the wild but extremely long-lived, making it a good landscape plant.

▼



circumstances. It is also worth remembering that large (and therefore expensive) plants are high risk – they are more prone to wind damage and drying out, and so large initial investments are more likely to be lost than small ones. Small plants are not only more sustainable, but a safer investment too!

NATIVES AND EXOTICS – THE DEBATE CONTINUES

The long-standing debate over the role of native and exotic (introduced) species continues, with an unfortunate tendency toward adopting entrenched positions in some countries (such as the USA) or arousing little interest in others (Japan). The key issue is the role of plants in gardens and designed landscapes to contribute to biodiversity by supporting food webs of insects, birds and other wild animals. It is worth noting the positions adopted by leading practitioners working in planting design. There does seem to be a consensus that using only native species is entirely appropriate in certain environments – chiefly rural ones or where the conservation of local and indigenous biodiversity is a priority. In many other situations, planting in the past would have used mostly or entirely non-native species, but now involves a larger proportion of natives.

The strictly nativist lobby, which believes in the exclusive use of regional natives, is not one which has arisen from within the garden and landscape communities, but rather has been imported from outside, from the world of environmentalist politics, where ecology is a word which can often weave dangerously between the evidence-based and scientific and the emotive and ideological. Unfortunately, the native plant lobby has on occasion acquired sufficient political support in some communities that the use of native plants has been mandated for landscaping projects, compromising their visual effectiveness and therefore the public support they receive.

Piet Oudolf's work at the Lurie Garden in Chicago and the New York High Line provides an example of synthesis which many in the profession would agree with. He chooses plants because they perform a function and meet certain visual criteria. In these two projects over half the plants used are natives of their region. One of the criteria for both commissions was that the

planting should reflect something of the regional natural environment: in the case of the High Line, this being the plant community which had established on the elevated rail line prior to restoration. A large proportion of natives was therefore essential. Crucially, though, the plants selected highlight just how good in design terms many of the plants in these regional floras are – a great number of species with garden and landscape potential have been largely ignored up until now. This is perhaps the most crucial point here – that prior to the current wave of interest in native plants, the nursery industry produced and sold what was beginning to look like a global flora of easy to use, easy to propagate plants. In the case of plants used by the landscape industry, they may have been different from one climate to another, but the effect was the same – too often both architecture and planting could be anywhere. The use of a proportion of locally native plants can do much to add a distinct signature to projects.

The native/exotic debate is complex, and here we will consider a range of points, some of which are more relevant to some localities than others.

- The issues are different from place to place, often very different. Take two island groups: the British Isles and New Zealand. The former has a very limited flora, with whatever managed to get across after the last ice age before the land bridge with the mainland was flooded by rising sea levels; the plant-eating invertebrate fauna at the base of the food web is largely generalist, and relatively few are totally dependent on native plant species. The local grass flora has a tendency to dominate – hugely reducing the ability of introduced species to spread. New Zealand also has a limited flora and fauna but one which has evolved in isolation and is extremely vulnerable to introduced plants, many of which have become invasive (the native flora possibly lacking its own vigorous pioneer species).

- Native plants are often a hugely underexploited design resource. The globalized garden and landscape flora alluded to above offers limited options. A walk in any natural habitat will reveal plants with ornamental or amenity potential, but they need to be brought into cultivation and evaluated for their visual and commercial potential, which takes time and effort. Increasingly, gardeners, nurseries and designers are realizing this, and undertaking this 'close to home' plant hunting. Even in



A meadow of the US native grass *Sporobolus heterolepis* in a garden on Nantucket Island, Massachusetts, includes *Echinacea purpurea*, a species which has become something of a poster boy for promoting native plants. It is important that planting schemes for biodiversity combine species which really support wildlife effectively as well as those which simply look good and tick the 'native' box.

Britain, with its restricted flora and long garden history, we are still learning to exploit the ornamental potential of our wildflowers – look at *Stachys officinalis*, almost unknown as an ornamental twenty years ago.

- Native plants enhance a sense of regional distinctiveness. Cultures sometimes want to emphasize their superiority by using 'civilized' plants. Hence the scandal when, in one of the rare iconic single actions in garden and landscape history, Roberto Burle Marx planted local plants in a public square in Recife, Brazil, in 1945. Nowadays the zeitgeist has moved toward celebration of the local and the regionally distinctive, about what sets a place apart and gives it its identity in the world. Locally native plants clearly have an important role to play here.

- The idea that exotic equals potentially invasive has no factual basis. What makes a plant an invasive alien is a question which ecologists continue to argue about. The reality is that of all the plants introduced into one country from another, only a few go feral and spread. The more dogmatic plant-nativists tend to tar all exotics with the same brush. Having said this, the nursery trade does have a responsibility to assess new plants for their ability to spread.

- Some native plants play a crucial role in local biodiversity webs, but this does not mean that exotic plants do not have any wildlife value. The food web which supports larger animals (mostly birds) is founded on invertebrates, chiefly insects. In many regions these are predominantly specialists – their larvae will *only* eat particular plants; a garden of introduced species will therefore support very few of them and thus greatly impoverish the food web. However, most other animals are more generalist: nectar-drinking insects such as bees are not tied to native species in this way, and neither are berry-eating birds. Adding species to a locality through

planting may actually improve the richness of the food web by providing nectar sources for bees at a time when the native flora has little to offer.

- Planting design is at least as important in supporting biodiversity as the species chosen. Research work undertaken in Britain at the BUGS project (Biodiversity in Urban Gardens) has shown that the most important factors in improving biodiversity are not the species used but the diversity of habitat. Trees, some shrubs, perennials, ground cover, and connectivity between different plant layers are what matter. And, a diversity of species.

- Planting design is, fundamentally, for people. In urban areas, planting in private gardens or public spaces is about providing a habitat for people. Anything which fails to interest or please them will lose support, as local governments that have created untidy 'wildlife areas' in parks have found out to their cost. In regions where poisonous spiders and snakes are common, there may be good reasons to actually fear such places! Areas for nature have to be seen to be attractive or in some way valued by people; only then will they gain political support. Using introduced species to provide interest for human users of landscape is one way to do this; very often non-gardening users of public spaces will find it easier to 'read' a planting if some familiar cultivated plants are included.

- There is plenty of room for both natives and exotics. Gardens, parks and areas of land around offices, malls, airports and road systems take up a lot of space on our crowded planet. Add up all the space which is currently mown, and which does not actually need to be 20 millimeter high grass, and the total will be huge. Globally it would probably be the equivalent of a modest-sized European country. There is plenty of space for both natives *and* exotics.



— TWO —

GROUPING PLANTS

How plants are grouped plays a major role in how they are seen and appreciated. It makes sense to start with what happens in nature, and then look at how plants have been grouped historically in gardens. Finally, we will consider the plant grouping systems which Piet Oudolf has used in his work since the mid-2000s.

NATURAL ENVIRONMENTS

When we look at a natural environment it is a snapshot in time. Come back ten years later and it could be very different. Despite the timeless feel to many natural environments, they are all in a state of constant change and flux – ecological science now tells us that there is no balance of nature, but only a constant ebb and flow of species. This is particularly true of the many environments we are familiar with which are not really natural but semi-natural, such as meadows, where an annual cut for hay prevents trees and shrubs from growing, or

even prairies, many of which were historically maintained by the burning practiced by Native Americans. Such environments are inherently unstable, and only constant human input maintains their particular range of plant life.

A good place to start is with an observation experiment. Look at a meadow (or indeed grassland managed for pasture) or prairie, and then compare it with a garden planting of perennials. What are the differences?

◀ A public park, Potters Fields in London.

GARDEN PLANTING	MEADOW OR PRAIRIE
Usually fewer than ten plants per square meter	Hundreds of plants per square meter
Usually one to five species per square meter	Up to 50 species per square meter
Individuals of a species often in groups	Individuals of species present, usually intensely intermingled
Almost all plant varieties present chosen for distinct aesthetic value	Plant community usually composed of dominant species (usually grasses or grass-like plants) – these species act as a matrix, together with smaller numbers of other species
Bare earth or ground-covering mulch often visible	Bare earth almost never present

We should point out here that meadows and prairies are maintained extensively – everything has to be treated as one, and there is no possibility of maintaining individual plants – whereas in a conventional garden or landscape planting, individuals are often treated differently. Thinking about these differences purely in design terms, what are the implications?

- A meadow or prairie lacks order, and plants often appear to be randomly distributed.
- A meadow lacks much graphic quality or structure, but instead appears fine-textured and diffuse, although a few prairie and dry meadow habitats contain species with a strong long-term structure as individuals.
- A meadow can look untidy after flowering, as it is almost impossible to maintain individual plants.
- A garden planting can only fit in so much to a unit area, which can make continual seasonal interest difficult.
- A garden planting can seem a restricted canvas, with little scope for repetition.

Clearly, there are advantages and disadvantages to the look of the meadow/prairie. Traditionally, garden design has largely disregarded the meadow as a garden feature, but more recent thinking has made many of us more aware of some of its aesthetic advantages. It is these which have encouraged a new generation of

gardeners and designers to reconsider the more diffuse beauty of the meadow and other grasslands.

Meadows and prairies are among many types of habitat where one species – or more usually one category – dominates, often grasses or grass-like plants, but with a large number of other species present in much smaller numbers. The whole forms a rich community, but in looking at it most of us tend to notice the highly visual minority element (the flowering perennials and occasional shrubs) rather than the dominant matrix of grasses which appear as a background. These type of habitats are actually very complex, with many species distributed through space with a high level of intermingling.

Thinking of other natural environments, what other types of plant grouping are there? At the opposite extreme to the meadow look are environments where one species appears to almost completely dominate. A good example is marshland colonized by reed (*Phalaris arundinacea*) or reedmace (cattails in North America, *Typha latifolia*), which are almost as much a monoculture as a field of maize or wheat. Between them are situations where plants form distinct patches, often intermingled with smaller species, for example in heathland with species of heather such as *Calluna vulgaris* and bilberry (*Vaccinium myrtillus*).



Astilbe 'Visions in Pink' is part of an intermingled plant mix on New York City's High Line. Repetition in a matrix of other plants evokes the repetition of plants in wild habitats.

PLANT GROUPING IN GARDEN HISTORY

During the nineteenth century summer bedding plants were grown in complex, often geometric and regular, patterns. At the turn of the nineteenth and twentieth centuries this style began to be applied to perennials as well, with the development of a style where plants were grown intermingled in strips, with each strip being repeated at regular intervals. Nowadays this is very rarely seen done with perennials, although it has been revived in France and Germany for temporary summer plantings dominated by annuals.

Planting in groups of a mass of individuals of the same species or cultivar dominated planting in the twentieth century – we shall refer to this as block planting. The British designer Gertrude Jekyll promoted the use of elongated blocks, called drifts, which had the effect of changing the way plants were seen as the viewer walked along. The Brazilian Roberto Burle Marx, an

artist by training, 'painted' with plants on a vast and dramatic scale, juxtaposing large blocks of strongly contrasting plants. Somewhat influenced by him have been the American partnership of James van Sweden and Wolfgang Oehme, who have also made effective use of large single-species blocks. For much of the century, however, block planting of a rather unquestioning and often lacklustre kind held sway, with countless landscape projects setting out plants, both perennials and small shrubs, in similar-sized blocks. Even in private gardens, block planting – at least where space allowed – held sway.

With the rise in interest in naturalistic planting, two developments have arisen which aim at a more thorough detailing of plant groupings. One is the randomization approach, which is derived from the almost-random effect of sowing a wildflower meadow from seed.

The other is the work of the German researchers Richard Hansen and Friedrich Stahl, who from the 1960s onwards developed a highly structured approach which aimed at a stylized representation of natural plant communities. Hansen and Stahl recognized five categories based on structural interest and level of grouping: theme plants, companion plants, solitary plants, ground-cover plants and scatter plants.

The selection of plants for Piet Oudolf's planting style has similarities to the Hansen and Stahl style. In particular, both he and they recommend that plantings comprise about 70 percent structural plants (those which maintain distinct visual structure for most of the growing season) and 30 percent filler plants (often rather formless, grown mostly for early season color).

The Hansen and Stahl approach is extremely useful, but is in danger of becoming formulaic. What is distinctive about the Oudolf design style is its constant evolution, at the core of which is now the intermingling of plant varieties.

A reaction against single-species block planting began around the end of the twentieth century. The movement toward planting for biodiversity was expressed through a growing interest in sowing wildflower mixes and in ecology more generally. In the UK, Netherlands and Germany more sophisticated and naturalistic approaches to putting perennials together were established. When we talk about breaking the rules, the rule that individuals of each variety have to be clumped together in blocks is the first one to be broken.

WOODY PLANTS

Conventional planting has relied very heavily on woody plants – not surprisingly, since they make a major impact on the landscape and the majority are very long-lived. In conjunction with perennials, they will inevitably change the growing conditions beneath them, making the ground layer suitable only for shade-tolerant species. The mixed border, beloved by British amateur gardeners, is a small-scale example of the possibilities for combining shrubs and perennials (and indeed also annuals, bulbs and climbers), but given its scale, and the

fact that these borders are often against a backdrop, it does often rather mean that the shrubs dominate – visually and ecologically. Larger scale and visually more innovative combinations of shrubs or small trees and perennials may allow more space for perennials.

One simple innovation which can make a great impact on many different scales is the shaping of hedges. Instead of cutting hedges straight, the individual plants in them can be given curves, so that each one stands out as an individual. This works particularly well with

ABILITY TO REGENERATE					
NON-EXISTENT → LIMITED → INCREASINGLY PROLIFIC → SUCKERING					
<i>Picea</i> spruce	<i>Quercus</i> oak	<i>Salix</i> willow	<i>Corylus avellana</i> hazel	<i>Aesculus parviflora</i> bottlebrush buckeye	<i>Rhus</i> sumac

The ability of woody plants to regenerate from cutting lies on a gradient, at one end of which there is no ability to regenerate from cutting back to the base, and at the other a strong tendency to produce suckers without any cutting back of the main trunk.



Large shrubs can be kept to a reasonable size by occasional cutting back to the base (coppicing). Frequent cutting back stimulates suckering, which can create an interesting effect. This technique can be applied to some familiar plants such as *Cotinus coggygria* (shown here) and the staghorn sumac (*Rhus typhina*) to limit their height and encourage suckers. It is, however, easy to pull out and keep the forest down to a small colony of variously sized individuals. It is used on the High Line, and since this plant is so familiar to semi-natural woodland edge habitats, including North American highway verges, its appearance can be a very effective way of saying 'natural', even two stories up in New York City.

mixed hedges, where the individuality given to the plant by its own cut is reinforced by its size relative to the others and its characteristic foliage color and texture. Hedges cut like this help to merge it, and the garden it contains, into wider rural landscapes, or indeed any landscape where trees are part of the wider view.

REGENERATION CUTTING

Trees and shrubs vary widely in their ability to regenerate, which has implications for their management and their design use. Cutting a tree down, for example, may not be the end of the tree; this will kill nearly all conifers, but most deciduous trees will grow back with multiple shoots, a technique which is made use of in the traditional woodland management technique of coppicing. Many shrubs regenerate themselves constantly from the

base – as old stems become senescent and deteriorate, vigorous young shoots emerge, growing straight up and eventually replacing the previous generations. The result is often a formless and tangled mass, but cutting back to base will result in a cleaner, more upright shape developing.

Some woody plants can spread through underground runners like many perennials, an example being the North American bottlebrush buckeye (*Aesculus parviflora*), forming large clumps. Species of *Rhus* are notorious among gardeners for suckering, but only after a period of existing as a deceptively tame-looking single-trunk tree. By cutting back every year or every other year, *Rhus* can be kept relatively small, sending up odd suckering shoots and creating the impression of a woodland-edge habitat with young trees merging with perennials and ground-covering plants.

A HIERARCHY OF PLANTING: PRIMARY PLANTS, MATRIX PLANTS AND SCATTER PLANTS

Take a quick look at a wild plant community such as a meadow, and you soon realize that you are not really seeing the plants. Our eyes tend first of all to be drawn to brightly colored flowers, and then to strong structures. The longer we look the more we see: more subtle colors, interesting shapes, juxtapositions and combinations. Repetition of key elements makes a big impact, with low-key features more likely to break through into our consciousness if they are extensively repeated – who notices a single white daisy in a field? But if it is repeated 100,000 times, then it will dominate all else. There is a good case for recognizing the importance of 'immediate impact' species in a planting – those which stand out and make the most impact.

Still looking at our wild plant community in early to mid-summer, if we try to disregard the most obvious plants, what else is there? There are quietly colored species, often with cream or buff flowers, and of course plants which do not have either strong colors or distinct structures, such as grasses. Finally, there is the inevitable fact that the vast majority of what we are looking at is foliage: green, undistinguished, background.

Compare a natural plant community with a garden planting. The latter is almost inevitably going to have more concentrated visual impact. But how much more? And how is this visual impact distributed, and what relationship does it have to its background? On looking at the design of gardens down the ages (excluding parks or wider landscapes), it would appear that there has been a long-term trend from high visual intensity to lower intensity, but also a more graduated range of impact on the eye. Modern recreations of Victorian bedding schemes can leave some onlookers with a headache: so many colors, so much going on, everything there for impact, with a consequent rapid onset of visual exhaustion. The perennial borders which followed them in the early twentieth century were also packed with intense colors and striking juxtapositions. As the century progressed, some gardeners began to try to promote plants which were more subtle: in Germany Karl Foerster (1874–1940) began to use grasses and ferns, while in England the artist Cedric Morris (1889–1982) surprised visitors to his

garden with his wide range of 'unconventional' plants. One of Morris's circle was the young Beth Chatto (born 1923), who in the 1960s puzzled some in the gardening establishment and inspired others with her choice of plants, chosen for form and line and natural elegance rather than eye-grabbing color: creamy *astrantias*, green *euphorbias* and broad-leaved *brunneras*.

Contemporary tastes in planting owe much to pioneers such as Chatto, but also to those who have promoted the creation of natural or semi-natural habitats as beautiful and worthwhile garden features: wildflower meadows, prairies and informal country hedgerows. Now, very much under the influence of the wildflower enthusiasts, gardeners and designers are more open to plantings where higher impact plants are set in a matrix of lower impact ones – just as the wildflowers in a meadow are only a small proportion of what is actually dominant: the grass matrix.

It is useful to think about a hierarchy of plants in terms of impact.

Primary plants are those which have the bulk of the impact – in conventional plantings all the plants can be considered as primary plants, although among them there will also be a clear hierarchy of impact. In a traditional English-style border, for example, there will be high-impact plants chosen for strong colors or structure being played off against lower impact ones such as pale lime-green *Alchemilla mollis*. For those new style plantings where higher impact plants are contrasted against a backdrop of lower impact ones, we will use the term 'matrix plants' for these quieter elements.

Matrix planting is where one or a limited number of plant species is used *en masse*, within which are embedded individuals or small to medium size groups of other, usually more visually prominent species.

A good analogy for the primary plant/matrix plant relationship is with a fruitcake, where nuts and fruit are scattered through a dough matrix. Having primary plants distributed through a matrix of others with lesser visual impact creates a more naturalistic and less obviously deliberate effect. It recalls, even if only psychologically, the visual feel of wild habitats, where a large

mass of low-impact plants are studded with stronger impact clumps or individuals.

Scatter plants are often most effectively added to a planting by literally randomly scattering them, for their role is primarily that of adding a sense of naturalness and spontaneity, and by being scattered throughout the whole, a sense of visual unity.

So, now we will look in more detail at what role plants in these categories might play in planting design and what kind of plants might be used. Unless specified, all schemes are Piet Oudolf's.

PRIMARY PLANTS – GROUPS

For many years, planting design on any scale larger than that of a domestic border involved putting plants in groups. These might be as small as three or involve blocks of hundreds of plants – the key point being that the group is a monoculture of one species or cultivar. There are some advantages to using single-variety groups of perennials, but these only really apply to larger public plantings and not to private gardens. One is that it simplifies maintenance where staff are unskilled

and cannot be adequately supervised. Another is that it allows the clear expression of leaf textures which can get lost in more diffuse plantings.

In public spaces there is also an educational reason for block planting; there is no doubting the role of parks and public gardens with a wide variety of perennials in inspiring the public to be more adventurous in using them in their own gardens, particularly in regions where there is little history of using perennials. In order to inspire people, they need to be shown how plants look, which is most easily achieved with single-variety groups. The impact of the plant's color and form can be seen right away, without viewers having to work out which flower belongs to which bit of foliage or how it relates to the ground beneath it. The character of a plant, that indefinable sum of habit, foliage form and flower color, can often only be really appreciated when it is grown in a single-species block – though if it is rangy or has unattractively sprawling stems when it looks at its best, then perhaps it would do better intertwined more naturalistically with its neighbors. Educational – but disadvantageous for the overall effect – can be the untidy or unattractive appearance of certain plants after



Here, in August at Pondsborne Nature Reserve in Norfolk (1996), the impact is created by groups of plants where multiple individuals of a variety are clumped. The red is *Helianthus 'Rubinzwerg'*. Several pale pink *Persicaria amplexicaulis* 'Rosa' also make an impact.

✱ (pages 84–85)

Taken when part of the Oudolf garden in the Netherlands was run as a nursery, this photograph illustrates very clearly two approaches to structural planting. One is the formal arrangement of geometric clipped shapes – in this case *Pyrus ussuriensis* 'Pendula' (silver pear) – and the other is the scattering of grass *Columagrostis* 'Earl Foweraker'. The grass with its long season of interest can be distributed in plantings in many different ways, all to great effect. Here it can be regarded as a random scatter element. The small tree in the foreground is *Rhus typhina*, which is coppiced every few years to limit its size.





Small clumps of purple *Stachys officinalis* 'Hummels' make a dramatic splash against pale green *Sedum autumnale*, which is here used as a matrix plant on the High Line in New York in July. Note how the stachys is repeated.



flowering, or when under stress or pest or disease attack. Group planting can throw a harsh spotlight on to a plant's character and point to its shortcomings.

Group-based planting can be made a lot more interesting in several ways:

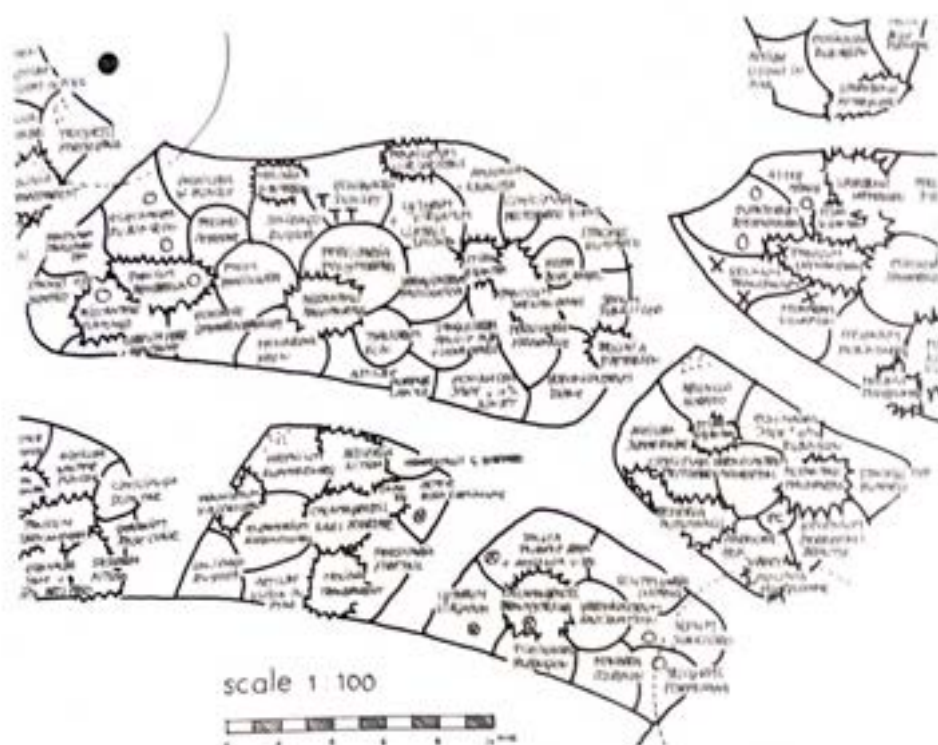
- Different sized groups can be used.
- Key groups can be repeated to develop a sense of rhythm.
- Different shapes of group can be used – as in the concept of the drift.
- Large groups can be interspersed with scattered repeating plants – either individuals or small groups.
- Groups can be made up of two or more different species, using various percentages of plants that combine well together or including a small percentage that flower earlier or later than the majority of the group.

The last method is the most radical as it challenges the dogma of the monocultural block, and is the first, albeit cautious, step toward the idea of mixed plantings.

Just as in all but the smallest domestic gardens repetition provides a sense of rhythm and unity, in large-

scale plantings which use more or less equal-sized groups there is also a need for clear repetition. Without repetition, group-based plantings lack a sense of unity. Just as a domestic garden with one of everything becomes a botanical collection – or 'plantsman's garden' in the polite terminology of British guides to gardens open to the public – so does a larger planting offering nothing but chunks of disparate plants. Repeated groups work best if made up of species with very good structure over a long time, or a long flowering season combined with a reasonably good appearance when no longer in flower. Taking the Trentham planting as an example (see opposite), there would appear to be a bewildering number of varieties – there are around 120 – but since many of these are very closely related, the number of truly distinctive plants is more like 70.

The eleven most frequently used plants in the 'Floral Labyrinth' at Trentham (see the table opposite) illustrate the value of repeating species with a long season of interest, and the importance of looking beyond flowering as the only or even main source of interest.



The 'Floral Labyrinth' at Trentham, Staffordshire (2004–7). This planting, in the grounds of what was a major country house garden and is now a visitor destination in the English Midlands, occupies an area around 120 meters long and 50 meters wide, including wide grass paths and two central grass areas. It has an interesting place in the development of the Oudolf planting style, as it is based overwhelmingly on the equal-sized groups which he used in previous large projects, such as Pensthorpe Nature Reserve in Norfolk (1996) and Dream Park in Enköping, Sweden (2003), but includes some hints of the levels of added complexity which he started to develop in projects which came later.

The overwhelming preponderance of grouped perennials is broken up a little by a few groups being mixtures, such as *Lythrum virgatum* with *Liatris spicata* (both deep pink narrow spikes) – shown on this plan section. Not visible on this comparatively small excerpt are also some small groups of repeating plants scattered through, such as *Baptisia* 'Purple Smoke' (very good bushy structure from May to October) and *Astrantia major* 'Roma' (long season of pink flowers).

Seasonal Interest of the 'Floral Labyrinth' at Trentham

	Early summer	Mid-summer	Late summer	Early autumn	Late autumn	Winter
<i>Astrantia</i> 'Roma'						
<i>Baptisia</i> 'Purple Smoke'						
<i>Echinacea</i> 'Rubinglow'						
<i>Eupatorium</i> 'Riesenschirm'						
<i>Lythrum virgatum</i>						
<i>Molinia</i> 'Transparent'						
<i>Persicaria</i> 'Firedance'					or first frost	
<i>Phlox</i> 'Amazone'						
<i>Sesleria nitida</i>						
<i>Stipa gigantea</i>						
<i>Veronicastrum</i> 'Fascination'						



Flowering

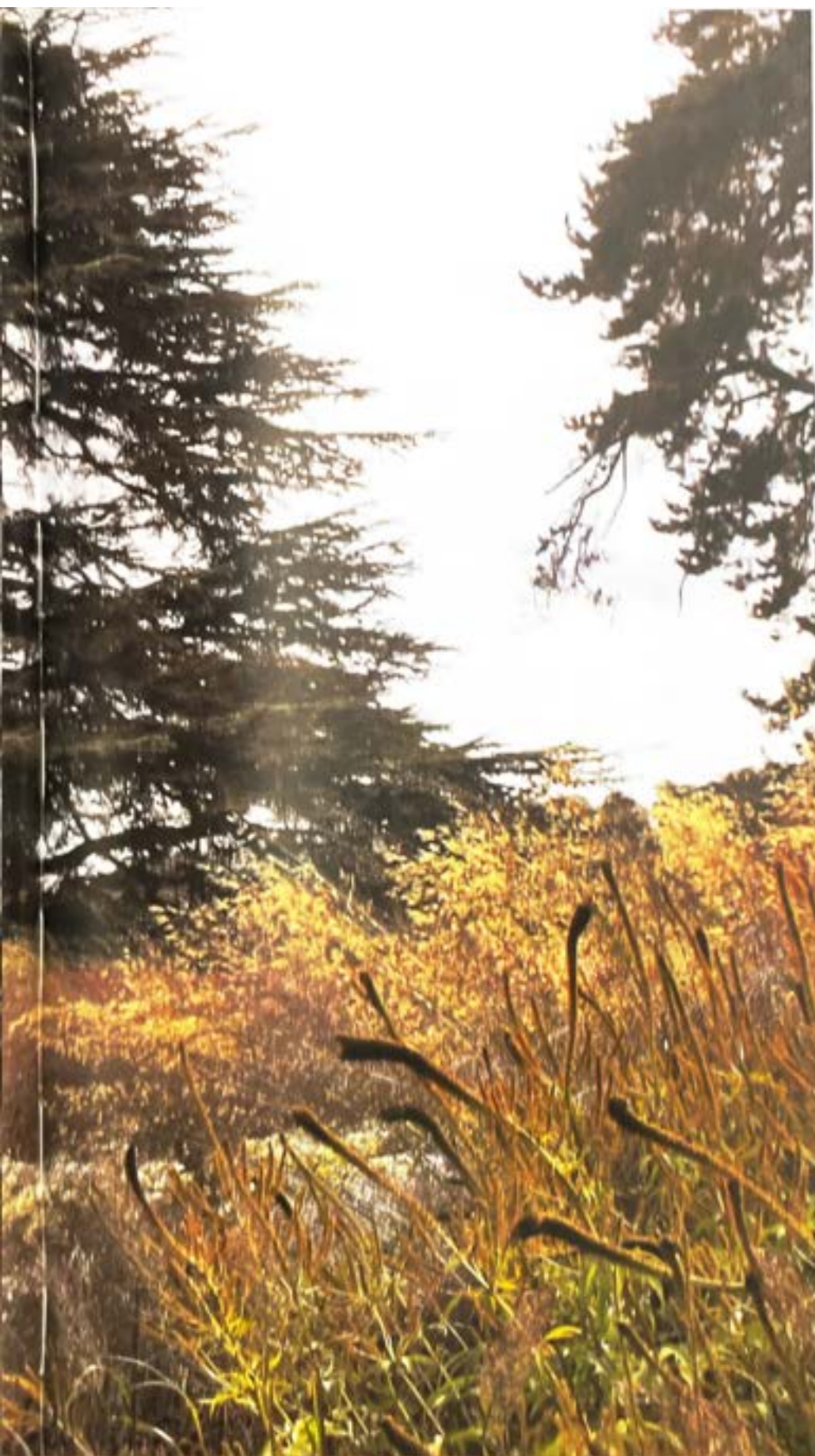


Foliage interest



Structural interest: seedheads, stems, grass flowers





Block planting at Trentham in the context of a historic landscape (the Cedar of Lebanon, *Cedrus libani*, would date to the eighteenth or early nineteenth century). Now, in September, the definition of grouped plants is partly lost and the area takes on a wilder look. The orange is *Heleum* 'Rubinweeg', while the grass on the right is *Molinia* 'transparent'. The odd dark 'deadlocked' seedheads in the right foreground are *Verbascum virginicum* 'Fascination'.

Repetition in block plantings

A large park such as Trentham planted in a very informal way with blocks of perennials in large beds among which it is possible to meander has a very definite similarity to what can be used on a smaller scale in the domestic garden with single plants or small groups. The principles are exactly the same. There should be a feeling of fluency so that as you walk through you are presented with not only variety (lots of different plants) but also the sense of familiarity which builds up when plants are repeated. Repetition can be regular, but in very informal plantings this has little meaning, so strutting for the random look is often better. Random repetition often works subliminally, so that as you walk around, the same species is seen again and again, not so often as to be obvious but often enough for it to work its way into the subconscious.

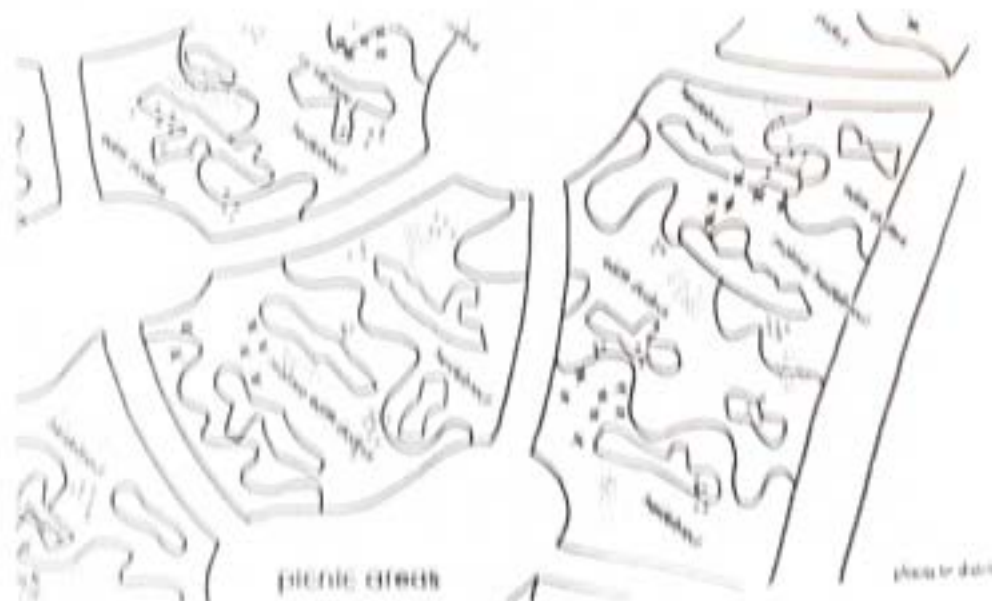
Repetition, of either individuals or blocks, works very differently as the year progresses. Early on, in spring or early summer when hardly any perennials are above half a meter high, repetition effects are obvious because everything is seen at once, and indeed are vital if the garden is not to seem disparate and messy. Later on at the height of summer and beyond, as perennial growth gets taller, the clarity of repetition becomes

much less obvious. If growth is so high that the exploration of a planting becomes a journey *through* rather than a look *over*, with vision restricted to what is growing immediately around, the experience becomes very different. Repetition will be appreciated by seeing the same plant or perhaps the same combination – in other words, it is appreciated as a function of repetition in *time* not repetition in *space*, and is more likely to be perceived subconsciously.

PRIMARY PLANTS – DRIFTS

Favored by Gertrude Jekyll, the drift is an easy way out of the dangers of the monotony which overreliance on block planting risks. Drifts are long and thin, and can be winding and snaking, so bringing different plants into close proximity. Often, given the tendency of many perennials to sprawl, trail and weave, there will be an impression of intermingling. More basically, however, the shape of the drift inevitably means that groups of plants will be seen differently from head-on to side-on, so there will be a sense of development as the viewer walks by. Jekyll's drifts were designed for the wide rectangular borders favored by early twentieth-century British garden architects.

At Trentham, two cultivars of grass *Molinia caerulea* are used to create what is essentially a matrix, into which are embedded perennials and some small shrubs. The grasses very much dominate, however, so the overall effect is that of a stylized meadow. What is important to stress here is that the two *Molinia* cultivars used ('Heidebraut' and 'Edith Dufresse') are both planted in commercial blocks, but in the form of complex drifts. If the two varieties had simply been mixed together the effect would have been a blur, as they are too similar to be easily distinguished. Planted as separate drifts, the distinction between the two can be appreciated, but the drift effect snails and intertwines them in a way which evokes the subtle patterns of grasses in natural grasslands.





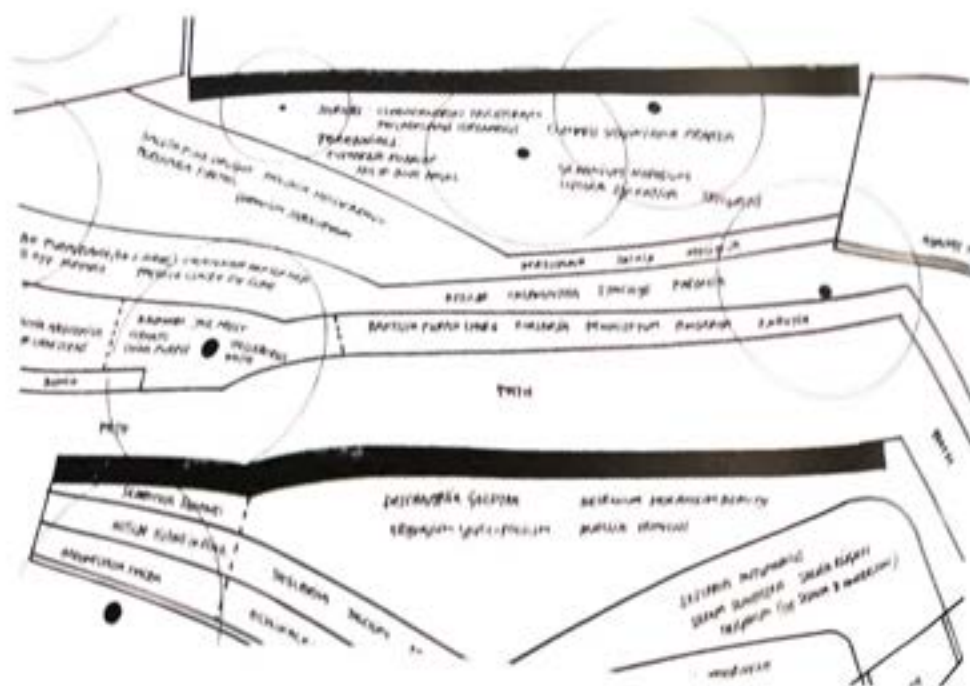
▲ Drifts of *Molinia caerulea* cultivars at Trentham in September.

► The rivers of grass at Trentham in early June with cultivars of *Iris sibirica* in flower among *Molinia caerulea* 'Heidebraut' and *M. c.* subsp. *caerulea* 'Edith Dudszus'. Some yellow *Trollius europaeus* and pale pink *Penicaria bistorta* are also visible – all are tolerant of the occasional flooding which affects this area.



◀ Rectangular drift planting at the Royal Horticultural Society garden at Wisley in Surrey (2001) uses strips of plant combinations of between four and six varieties seen long end on to viewers as they walk down a central grass aisle. The geometry of the plan is invisible beneath vigorous, intermingled plant growth. Here deep blue spikes of *Agastache* 'Blue Fortune' are intermingled with white *Eryngium yuccifolium*; in the background purple *Veronicastrum virginicum* 'Fascination' is visible in another strip. The shrub to the right is *Cotinus coggygria* with its distinctive seedheads.

A detail from the plan for Potters Fields Park, London (2007), where jagged drifts are an effective way of creating simple mixed planting combinations.



▲ At Potters Fields Park, London (2007), drifts of grass and perennial combinations create an orderly but dynamic effect. In the foreground is a mix of *Echinacea purpurea* and white-flowered, sprawling *Calamintha nepeta* subsp. *nepeta*. The grass in the background is *Deschampsia cespitosa* 'Goldtau'.



▲ A drift of *Sesleria autumnalis* in the foreground with red *Helenium* 'Moerheim Beauty' and *Deschampsia* behind. Other perennials are also included in each drift, but are not visible here. The use of drifts in this park creates a strong sense of movement and maximizes the trade-off between relatively simple, easy-to-maintain planting and visual complexity.

An advancement on drifts is the use of simple combinations of up to five or six varieties in a large drift. This was used extensively in the double borders at the RHS Wisley garden in Surrey in 2001, where each drift was exactly the same size and rigidly geometric. The reality on the ground is that plant foliage spills over the straight edges, and given that each drift is a mixture, there is no sense of regimentation.

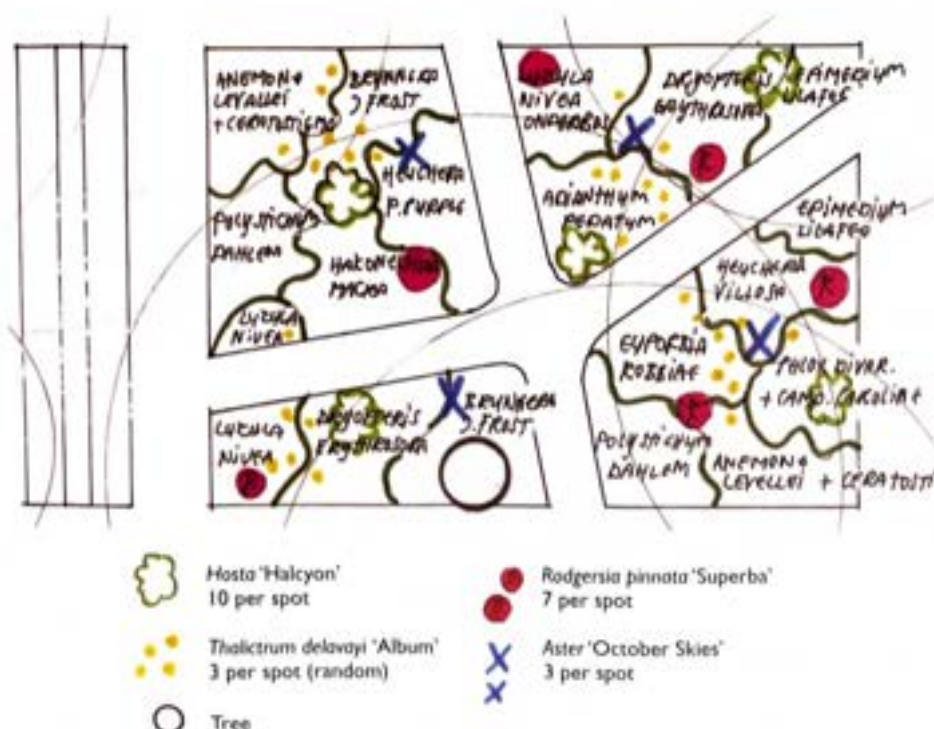
Drifts offer a good midway format for gardeners and designers who want to break away from traditional block plantings but feel that they do not have the experience to attempt more complex mixed plantings. Drifts can create an illusion of intermingling, and they have certain advantages for ongoing management above mixed plantings. If there are plants which need a mid-season tidy-up or pruning, it is easier to gain access to them as well as to carry out additional planting – for example of bulbs, which are notoriously difficult to add to existing perennial plantings. Perhaps most importantly, by keeping the complexity low and relatively pre-

dictable, it is easier for maintenance staff with low plant knowledge to weed and manage the planting.

REPEATING PLANTS

Repeating plants can be used individually or in small groups at – very generally – regular intervals to add rhythm and variation to block plantings and to generally break up their chunkiness. Fundamentally they are about creating a sense of unity; whether in a private garden or a large public space, the repetition of a few distinct long-season plants creates a feeling that 'this is one place, with one design and one vision'. They can be used to lead the eye and so direct the viewer.

On a smaller scale, one particular area can be given a sense of unity with repeating plants, and in this case it will be more about stamping a distinct personality on that area with respect to the rest of the garden. An example of both these uses of repeating plants can be seen in the plans for Leuvehoofd (see pages 107–109).



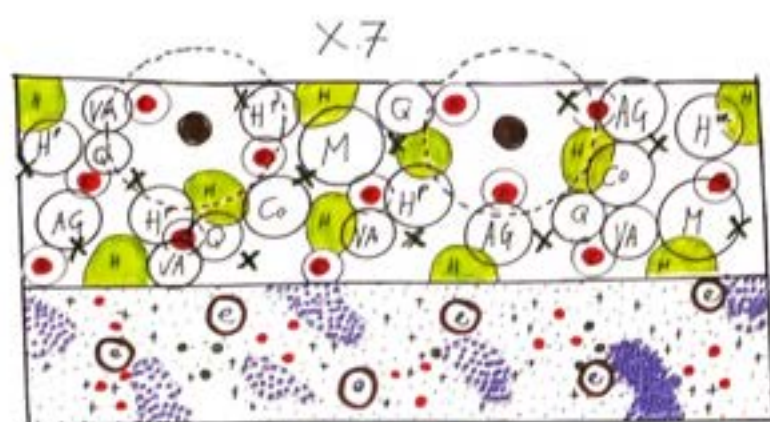
A shaded area from the van Veggel garden, Netherlands (2011), the large pencil arcs representing tree canopies. This excerpt from a much larger plan illustrates group planting of shade-tolerant plants interspersed with repeating plants. The key below (left) is for the repeating plants. Numbers for the repeating plants are given for each mark, with an additional annotation for *Thalictrum delavayi* 'Album' that they be scattered. All the repeating plants used are quite solid clump-formers, with the *Thalictrum* being an exception. Its tall stems and fluffy flower-heads are relatively insubstantial, and so small groups of it need to be repeated in order to make much of an impact. Note the wavy edges to the blocks – this blurs the boundaries between plant groups.

- (M) *Magnolia x soulangeana* 'Solitaire'
 (Co) *Cornus kousa*
 (Ag) *Acer griseum*

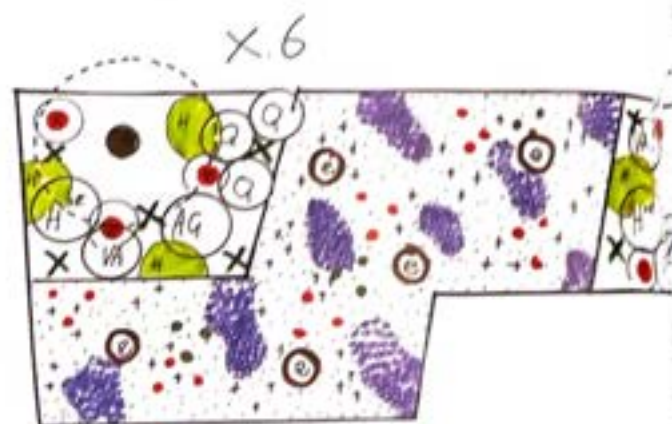
- (H^p) *Hamamelis x intermedia* 'Pallida'
 (H^{op}) *Hamamelis x intermedia* 'Orange Peel'
 (Vb) *Viburnum x rhytidophyloides* 'Alleghany'
 (Q) *Hydrangea quercifolia* 'Flemingia'

- Iris sibirica* 'Perry's Blue'
 (E) *Eupatorium maculatum* 'Purple Bush'
 (A) *Aster x frikartii* 'Mönch'
 (An) *Anemone x hybrida* 'Pamina'
 (P) *Peucedanum verticillare*

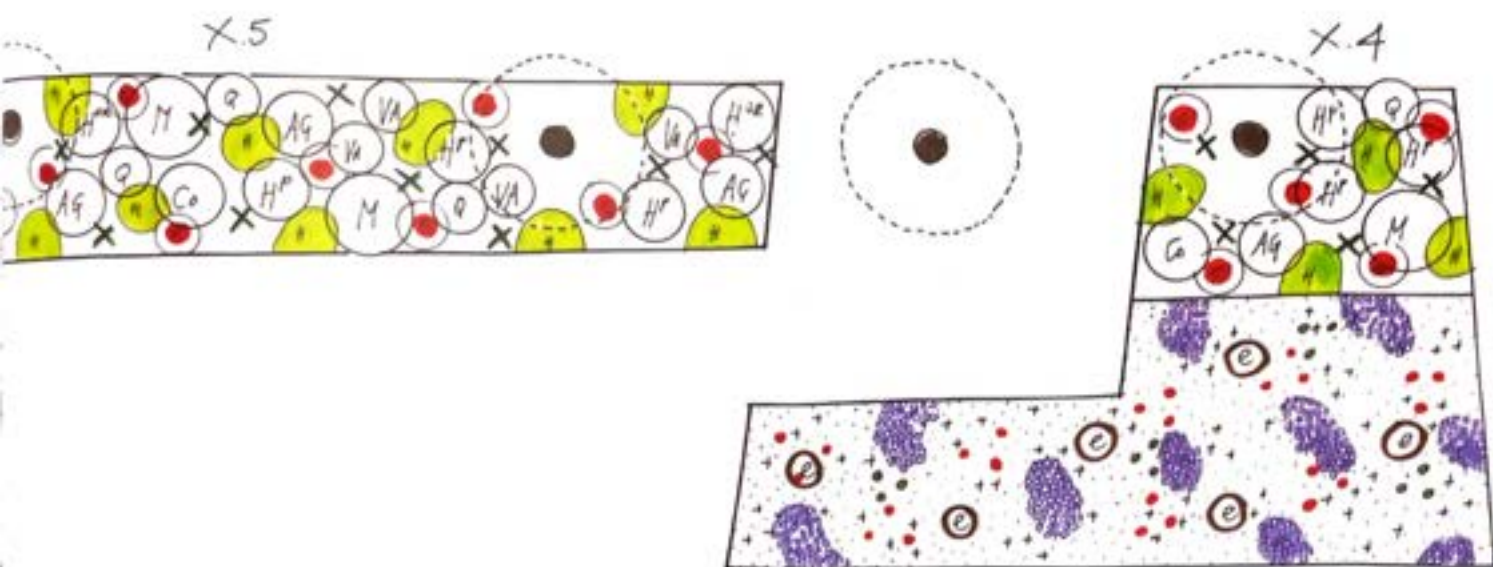
Fill up with:
Malva caerulea subsp. *caerulea* 'Edith Dudsza'



Client: DS+V Gemeente Rotterdam
 Planting Design Westerkade
 Scale: 1:100
 Date: 30 January 2010
 Design: Piet Oudolf, Hummelo



- (H) *Hakonechloa macro*
 (Ep) *Epimedium x perralchicum* 'Fröhneleiten' X.5, X.6, X.7
 (Eg) *E. grandiflorum* 'Lilafée' X.4
 (X) *Polystichum setiferum* 'Herrenhausen'
- Fill up with: *Geranium x oxonianum* 'Claridge Druce' 60%
Heuchera villosa 'Brownies' 30%
Asperula odorata 10%



A public planting for the Westerkade – the old quayside of the river Maas in Rotterdam, where beds of varying size run for several hundred meters. There are two planting combinations. One is around the canopy of some existing elm trees and uses small ornamental trees and shrubs such as varieties of viburnum and hydrangea and perennial underplanting for foliage interest. The other is for

the more open areas and uses a matrix of a molinia grass and a limited range of perennials. Of these, the tall, self-seeding umbellifer *Prucedanum verticillare*, was intended to provide an element of the unpredictable as well as an imposing winter sight. There were, however, not enough *Prucedanum* so some of the smaller, dark-flowered *Angelica gigas* were also used.



Excerpt from a plan for the Maximilianpark in Hamm, Germany (2009–10).

Good repeating plants need to have a distinct personality and a long season of interest, or at least disappear tidily or die back discretely, as seen in the plan from the van Veggel garden in the Netherlands on the previous pages:

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- *Hosta* (Tardiana Group) 'Halcyon' – looks good from spring to autumn.
 - *Aster oblongifolius* 'October Skies' – tidy mounds, late flowers.
 - *Polystichum setiferum* 'Herrenhausen' – looks good from spring to autumn.
 - *Polygonatum xhybridum* 'Weihenstephan' – flower and structure in spring, can go into early dormancy but does so tidily.
 - *Tricyrtis formosana* – relatively inconspicuous until late flowering.
 - *Salvia pratensis* 'Pink Delight' – combines color and structure and looks tidy after flowering.
 - *Thalictrum delavayi* 'Album' – relatively inconspicuous before and after flowering. Its transparency takes the eye away from the stiffness and solidity of any surrounding groups.

At the Maximilianpark in Germany, groups of roughly similar size are interspersed with repeating plants, chosen for long-season color or structure interest or else late interest often preceded by attractive foliage. An exception might be *Geranium psilostemon*, which like all geraniums has poor structure and can look untidy after flowering, although it disappears among taller planting later in the season. It is, however, valuable for its vibrantly colored flowers in early summer, which have a great capacity to seize the attention and so stamp a unifying personality on a planting. It is interesting to look at the numbers of the repeating plant species in the plan. There are twice as many groups of the species which are used the most – that is, certain plants are used to visually dominate. These include two grasses (*Molinia* 'Transparent' and *Panicum* 'Shenandoah') – structural but at the same time relatively low-key – and *Aster tartaricus* 'Jindai', a late, bright variety with an upright

and unusually neat structure for an aster. Among the repeating plants with the lowest level of appearance are a number (such as *Persicaria amplexicaulis* 'Orange Field' and *Inula magnifica* 'Sonnestrahl') which are physically large and bulky – not many are needed to make an impact.

The numbers of the repeating plants used are as follows (see the table below):

- Three per circle for *Eupatorium* 'Chocolate', *Geranium psilostemon*, *Inula magnifica* 'Sonnestrahl', *Molinia* 'Transparent', *Panicum* 'Shenandoah' and *Sanguisorba tenuifolia* 'Alba'.
- Three to five per circle for *Aruncus* 'Horatio', *Persicaria amplexicaulis* 'Orange Field' and *Veronicastrum virginicum* 'Album'.
- Seven to nine per circle for *Aconitum* 'Spark's Variety', *Aster tartaricus* 'Jindai' and *Thalictrum rochebrunianum*.

Seasonal Interest at the Maximilianpark, Germany

	Early summer	Mid-summer	Late summer	Early autumn	Late autumn	Winter
<i>Aconitum</i> 'Spark's Variety'						
<i>Aruncus</i> 'Horatio'						
<i>Aster tartaricus</i> 'Jindai'						
<i>Eupatorium</i> 'Chocolate'						
<i>Geranium psilostemon</i>						
<i>Inula magnifica</i> 'Sonnestrahl'						
<i>Molinia</i> 'Transparent'						
<i>Panicum</i> 'Shenandoah'						
<i>Persicaria amplexicaulis</i> 'Orange Field'						
<i>Sanguisorba tenuifolia</i> 'Alba'						
<i>Thalictrum rochebrunianum</i>						
<i>Veronicastrum virginicum</i> 'Album'						



Flowering



Foliage interest



Structural interest: seedheads, stems, grass flowers



Yellow *Rudbeckia subtomentosa* is repeated along a stretch of the High Line, set in a matrix of grasses, mostly *Sporobolus heterolepis* and *Panicum 'Shenandoah'*, in August. This effect is remarkably similar to the grass-dominated and perennial flower-studded semi-natural habitat to be seen alongside many American highways and in prairie remnants.

MATRIX PLANTING

The concept of matrix planting is one which has been around for some time, but like many terms in planting design it has had the unfortunate history of being used by different people to mean different things – in part because of a misunderstanding over the meaning of the word. 'Matrix' is defined in the *American Heritage Dictionary of the English Language* as 'a surrounding substance within which something else originates, develops, or is contained', and it is with this in mind that we will use the term – we have already suggested the analogy of a fruitcake for describing a matrix.

The matrix evokes the situation in many natural habitats where a small number of species form the vast majority of the biomass, studded with a larger number of species present in much smaller numbers but which are a visually important element. Consequently, good matrix plants are visually quiet, with soft colors and without striking form. They also need to be effective in physically filling space – part of their function is to be a ground cover (or at least to hide the soil surface), so they need to mesh together well. It is vital that they always look good or at least acceptably tidy, so that even when they have finished their main season of interest they have good structure, do not flop over or look sad and bedraggled.

Grasses are the obvious plants to use for matrix planting, especially tussock-forming or bunch grasses (that is, cespitose species), or the denser and slower growing clump-forming species. For a variety of reasons to do with their physiological efficiency, grasses dominate most open temperate zone habitats. Researchers and practitioners like James Hitchmough and Cassian Schmidt have suggested that using cespitose grasses might be a good way to create low-maintenance plantings. Many such grasses are very long-lived and stable, with 'closed nutrient cycles': that is, they recycle nutrients from their old leaves as they fall down and rot around them. The environment they create is potentially a competitive one, which would have the desirable effect of reducing weed growth. Yet because they do not form runners and so create a complete canopy in the way that turf grasses or slowly spreading ones like *Miscanthus* or *Calamagrostis* 'Karl Foerster' do, they will

not compete so strongly with flowering perennials that they risk eliminating them. Long-lived perennials or other species which have a similar form of growth may be considered as stable companions in the long term. Early Oudolf matrix-type plantings used *Deschampsia cespitosa* as the dominant grass; in cultivation on fertile soils it is relatively short-lived, but it can also self-sow strongly, so the chances of the long-term development being either no-*Deschampsia* or too-much-*Deschampsia* are all too likely. The good thing is that *Deschampsia* does appear to be compatible with a great many other plants – it does not totally out-compete them. Varieties of *Molinia caerulea*, a long-lived species, also offer the hope of greater stability, although the tight clumps of many cultivars of these plants leave a lot of bare soil between them and so are best in combination with low spreading or sprawling perennials such as varieties of *Calamintha*. *Sporobolus heterolepis* has great potential as it has the fuzzy charm of *Deschampsia cespitosa* and is an important natural matrix plant; in its native dry prairie it is known to live for decades; it is, however, slow to establish in cooler European climates.

The potential of *Carex* (sedges) and other plants which can substitute for grasses both ecologically (in that they naturally dominate) and visually (they have long narrow leaves) to act as matrix plants is enormous; there is a feeling that the surface has hardly been scratched with these long-lived, often evergreen, stress-tolerant and resilient plants. Other plant genera which offer similar evergreen possibilities include *Luzula* in cool, moist climates and *Liriope* and *Ophiopogon* in climates with warm, humid summers – the latter are commonly used in the south-east USA, much of eastern China and Japan. Behavior varies from the strongly spreading to the tight clump, so huge opportunities for gardeners and landscape designers exist here. All these plants together are commonly known as 'grass-like plants' – it is important to understand that although they may look like grasses, they are not; grasses have a different physiology and tend to be greedier for light and nutrients.

Other potential matrix plants include clump-forming species grown chiefly for their foliage, like varieties of *Heuchera*, *Tellima*, *Epimedium* or woodland species

A simple matrix planting at Bury Court, Hampshire, with *Molinia caerulea* forming a meadow-like mass in which *Digitalis ferruginea*, a short-lived but self-seeding foxglove, and *Allium sphaerocephalon* are flowering in July.



of *Saxifraga*. Plants such as these, or with similar semi-evergreen leaves and a spreading habit, often form a major component of the ground flora in woodland. *Iris sibirica* is possible as a lesser component in matrix planting – its flowering season is so short that its strap-like leaves make it function almost as a grass, it always looks tidy, is very long-lived and has good seedheads; a habit of crushing competitors with a dense thatch of slow-to-rot leaves does, however, mean that it can only be combined with equally robust plants or that a maintenance program specifies their removal.

Finally, there is the potential of certain late flowering perennials as minor components in a matrix. *Limonium platyphyllum*, those varieties of *Sedum* descended from *S. spectabile* or *S. telephium*, and several species of *Eryngium* (for example, *E. bourgatii*) are very long-lived, stress-tolerant – particularly of drought – and never look untidy. In Chicago's Lurie Garden, the broad heads of *Limonium platyphyllum*, composed of thousands of tiny flowers, form a soft haze that other flowers blend and blur into, showing the potential of the visual aspect of the matrix.

One thing which needs to be borne in mind about the limited range of plants used in matrix-based plantings is that they are relatively stable in the sense that they continue to occupy the same space over long periods of time. This is true of many caespitose grasses, but also of flowering perennials like *Sedum telephium* varieties and *Limonium platyphyllum*. *Iris sibirica* and the low clump formers like *Tellima* do spread slowly, but they have a proven ability – in suitable conditions, of course – to dominate and hold space over long periods.

While the use of caespitose grasses, and sedges as well, is clearly an application of what is often seen in nature, the use of non-grass clump formers also has a natural model in that some woodland-floor habitats are dominated by these plants, at least where light and other conditions allow. Among clump-forming perennials, though, there is considerable variation in medium- to long-term performance, even between cultivars within the same genus. This is emphasized because of the importance of selecting plants which will do the functional ground-covering work demanded of them. In the case of *Heuchera*, for example, some cultivars do this well